

[MS-RAIW]:

Remote Administrative Interface: WINS

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Revision Summary

Date	Revision History	Revision Class	Comments
10/24/2008	1.0	New	Version 1.0 release
12/5/2008	1.1	Minor	Clarified the meaning of the technical content.
1/16/2009	2.0	Major	Updated and revised the technical content.
2/27/2009	2.0.1	Editorial	Changed language and formatting in the technical content.
4/10/2009	3.0	Major	Updated and revised the technical content.
5/22/2009	4.0	Major	Updated and revised the technical content.
7/2/2009	4.0.1	Editorial	Changed language and formatting in the technical content.
8/14/2009	4.0.2	Editorial	Changed language and formatting in the technical content.
9/25/2009	4.1	Minor	Clarified the meaning of the technical content.
11/6/2009	4.1.1	Editorial	Changed language and formatting in the technical content.
12/18/2009	4.1.2	Editorial	Changed language and formatting in the technical content.
1/29/2010	4.2	Minor	Clarified the meaning of the technical content.
2/11/2011	5.0	Major	Updated and revised the technical content.
3/25/2011	5.0	None	No changes to the meaning, language, or formatting of the technical content.
5/6/2011	5.1	Minor	Clarified the meaning of the technical content.
6/17/2011	5.2	Minor	Clarified the meaning of the technical content.
9/23/2011	5.2	None	No changes to the meaning, language, or formatting of the technical content.
12/16/2011	6.0	Major	Updated and revised the technical content.
3/30/2012	7.0	Major	Updated and revised the technical content.
7/12/2012	7.1	Minor	Clarified the meaning of the technical content.
10/25/2012	7.1	None	No changes to the meaning, language, or formatting of the technical content.
1/31/2013	7.1	None	No changes to the meaning, language, or formatting of the technical content.
8/8/2013	8.0	Major	Updated and revised the technical content.
11/14/2013	8.0	None	No changes to the meaning, language, or formatting of the technical content.
2/13/2014	8.0	None	No changes to the meaning, language, or formatting of the technical content.
5/15/2014	8.0	None	No changes to the meaning, language, or formatting of the technical content.
6/30/2015	9.0	Major	Significantly changed the technical content.

Date	Revision History	Revision Class	Comments
10/16/2015	9.0	None	No changes to the meaning, language, or formatting of the technical content.
7/14/2016	9.0	None	No changes to the meaning, language, or formatting of the technical content.
6/1/2017	9.0	None	No changes to the meaning, language, or formatting of the technical content.
9/15/2017	10.0	Major	Significantly changed the technical content.
9/12/2018	11.0	Major	Significantly changed the technical content.
4/7/2021	12.0	Major	Significantly changed the technical content.
6/25/2021	13.0	Major	Significantly changed the technical content.
4/23/2024	14.0	Major	Significantly changed the technical content.

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1 Introduction

This is a specification of the Remote Administrative Interface: WINS protocol. This protocol defines **remote procedure call (RPC)** interfaces that provide methods for remotely accessing and administering a server for the **Windows Internet Name Service (WINS)**. This protocol is a client/server protocol that is based on RPC and is used in the configuration, management, and monitoring of a **WINS server**.

An application implementing this protocol can remotely perform service monitoring of a WINS server as well as creating, updating, querying, or deleting database records, performing database **scavenging**, and **replicating** the database records with other WINS servers.

Sections 1.5, 1.8, 1.9, 2, and 3 of this specification are normative. All other sections and examples in this specification are informative.

1.1 Glossary

This document uses the following terms:

active: A state of an attributeSchema or classSchema object that represents part of the schema. It is possible to instantiate an active attribute or an active class. The opposite term is defunct.

active record: A **name record** that has been registered but not released.

address version map: See **owner version map**.

authentication level: A numeric value indicating the level of authentication or message protection that **remote procedure call (RPC)** will apply to a specific message exchange. For more information, see [\[C706\]](#) section 13.1.2.1 and [\[MS-RPCE\]](#).

browser name: A **NetBIOS name** whose 16th character is set to 0x1B. This name is used to identify the domain master browser server for a domain.

client: A computer on which the remote procedure call (RPC) client is executing.

dynamic record: A **name record** that is created through NetBT name registration by a **client**.

endpoint: A network-specific address of a remote procedure call (RPC) server process for remote procedure calls. The actual name and type of the endpoint depends on the **RPC** protocol sequence that is being used. For example, for RPC over TCP (RPC Protocol Sequence ncacn_ip_tcp), an endpoint might be TCP port 1025. For RPC over Server Message Block (RPC Protocol Sequence ncacn_np), an endpoint might be the name of a **named pipe**. For more information, see [\[C706\]](#).

extinction interval: The interval at which released names are changed to the **tombstone state**.

h-node: NetBT h-node, also referred to as h-node or Hybrid node, is a combination of b-node and p-node functionality. H-node uses point-to-point communication first. If the NetBIOS name server cannot be located, it switches to broadcast. H-node continues to poll for the name server and returns to point-to-point communication when one becomes available.

HostName: The name of a host on a network. Users specify computers on a network by their host names.

Interface Definition Language (IDL): The International Standards Organization (ISO) standard language for specifying the interface for remote procedure calls. For more information, see [\[C706\]](#) section 4.

Internet Protocol version 4 (IPv4): An Internet protocol that has 32-bit source and destination addresses. IPv4 is the predecessor of IPv6.

Internet Protocol version 6 (IPv6): A revised version of the Internet Protocol (IP) designed to address growth on the Internet. Improvements include a 128-bit IP address size, expanded routing capabilities, and support for authentication and privacy.

IPv4 address in string format: A string representation of an **IPv4** address in dotted-decimal notation, as described in [\[RFC1123\]](#) section 2.1.

little-endian: Multiple-byte values that are byte-ordered with the least significant byte stored in the memory location with the lowest address.

multihomed: (1) Having two or more network interfaces on which NetBIOS over TCP is enabled.

(2) Multiple network interfaces to multiple separate physical networks and thus multiple IPv4 addresses.

multihomed machine name: The **NetBIOS name** of a machine that is **multihomed**.

name record: The **NetBIOS name**-to-**IPv4** address mapping.

name resolution: The process of resolving a **NetBIOS name** to an **IPv4** address.

named pipe: A named, one-way, or duplex pipe for communication between a pipe server and one or more pipe clients.

NBNS pull partner: A NetBIOS name server that requests new **NBNS name records** (replicas) from its **partner**.

NetBIOS: A particular network transport that is part of the LAN Manager protocol suite. **NetBIOS** uses a broadcast communication style that was applicable to early segmented local area networks. A protocol family including name resolution, datagram, and connection services. For more information, see [\[RFC1001\]](#) and [\[RFC1002\]](#).

NetBIOS name: A 16-byte address that is used to identify a **NetBIOS** resource on the network. For more information, see [\[RFC1001\]](#) and [\[RFC1002\]](#).

NetBIOS Name Server (NBNS): A server that stores NetBIOS name-to-IPv4 address mappings and that resolves NetBIOS names for NBT-enabled hosts. A server running the Windows Internet Name Service (WINS) is the Microsoft implementation of an NBNS.

NetBIOS scope: The population of computers across which a registered **NetBIOS name** is known. Each NetBIOS scope has a scope identifier, which is a character string that meets the requirements of the Domain Name System (DNS) for domain names.

NetBIOS suffix: The 16th byte of a 16-byte **NetBIOS name** that is constructed using the optional naming convention defined in [\[MS-NBTE\]](#) section 1.8.

NetBT b-node: A **NetBT node type** that is configured to use broadcast **NetBIOS name** queries for name registration and resolution.

NetBT h-node: A combination of **NetBT b-node** and **p-node** functionality. An h-node uses point-to-point communication first. If the **NBNS** cannot be located, h-node switches to broadcast. The h-node continues to poll for the name server and returns to point-to-point communication when one becomes available.

NetBT m-node: A **NetBT node type** that uses a mix of b-node and **p-node** communications to register and resolve **NetBIOS names**. An m-node uses broadcast resolution first; then, if necessary, it uses a server query.

NetBT node type: The transport mechanism used to resolve **NetBIOS names** that are broadcast, multicast, or unicast.

normal group: A group of hosts that does not have an associated address. It is assumed to be valid on any subnet.

owner: A security principal who has the requisite permission to manage a security group.

owner NBNS server: An **NBNS server** that handles the name registration of a **client** and so owns the mapping for that **client**. An owner NBNS server is also referred to by the term **owner WINS server**.

owner version map: A table in which each entry has two fields, owner and version number. The owner field contains a **WINS server** address; the version number field contains the highest version number of all the records owned by the **owner WINS server** that are stored at the local **WINS server**.

owner WINS server: See **owner NBNS server**.

partner: A computer connected to a local computer through either inbound or outbound connections.

p-node: When using p-node NetBIOS name resolution, broadcasts are not used for name registration or NetBIOS name resolution. Instead, all systems register themselves with a NetBIOS Name Server (NBNS) upon startup. The NBNS is responsible for mapping computer names to IPv4 addresses and making sure that no duplicate names are registered on the network.

priority class: An attribute of a process that is used to determine the scheduling priority of threads of that process. The priority of a thread is determined by a combination of the priority class of its process and the priority level of the thread within the priority class.

pull partner: See **NBNS pull partner**

RELEASED: The state of a **name record**, in which its name has been explicitly released through a name release request, or in which it has failed to be refreshed by name by a **client** within the renewal interval.

released record: A **name record** that has been explicitly released through a name release request; or a **name record** that a **client** has failed to refresh by name within the renewal interval.

remote procedure call (RPC): A communication protocol used primarily between client and server. The term has three definitions that are often used interchangeably: a runtime environment providing for communication facilities between computers (the RPC runtime); a set of request-and-response message exchanges between computers (the RPC exchange); and the single message from an RPC exchange (the RPC message). For more information, see [C706].

replication: The process of propagating the effects of all originating writes to any replica of a naming context (NC), to all replicas of the NC. If originating writes cease and replication continues, all replicas converge to a common application-visible state.

replication partner: See NBNS replication partner

scavenging: A regularly scheduled process in which the state of database records are changed if they have not been updated within a certain time interval, measured by the process that checks whether current time exceeds the record's time stamp value.

security support provider (SSP): A dynamic-link library (DLL) that implements the Security Support Provider Interface (SSPI) by making one or more security packages available to applications. Each security package provides mappings between an application's SSPI function

calls and an actual security model's functions. Security packages support security protocols such as Kerberos authentication and NTLM.

special group: A group of hosts that have a single name. When a name registration is received for a special group, the actual address rather than the limited broadcast address is stored in the group. When a name query is received for such a group, the **IPv4** addresses that have not timed out are returned.

static record: A manually created entry in the database of a **NBNS server**.

target WINS server: The **WINS server** on which the **RPC** method call is being executed.

tombstone: An individual record of scheduling data that represents a Meeting object where an attendee declined a meeting.

tombstone interval: See **extinction interval**.

tombstone state, tombstoned: The state of a **released record** that is not re-registered or refreshed by a **client** within the **extinction interval**.

Transmission Control Protocol (TCP): A protocol used with the Internet Protocol (IP) to send data in the form of message units between computers over the Internet. TCP handles keeping track of the individual units of data (called packets) that a message is divided into for efficient routing through the Internet.

Unicode: A character encoding standard developed by the Unicode Consortium that represents almost all of the written languages of the world. The **Unicode** standard [\[UNICODE5.0.0/2007\]](#) provides three forms (UTF-8, UTF-16, and UTF-32) and seven schemes (UTF-8, UTF-16, UTF-16 BE, UTF-16 LE, UTF-32, UTF-32 LE, and UTF-32 BE).

Unicode string: A **Unicode** 8-bit string is an ordered sequence of 8-bit units, a **Unicode** 16-bit string is an ordered sequence of 16-bit code units, and a **Unicode** 32-bit string is an ordered sequence of 32-bit code units. In some cases, it could be acceptable not to terminate with a terminating null character. Unless otherwise specified, all **Unicode strings** follow the **UTF-16LE** encoding scheme with no Byte Order Mark (BOM).

universally unique identifier (UUID): A 128-bit value. UUIDs can be used for multiple purposes, from tagging objects with an extremely short lifetime, to reliably identifying very persistent objects in cross-process communication such as client and server interfaces, manager entry-point vectors, and **RPC** objects. UUIDs are highly likely to be unique. UUIDs are also known as globally unique identifiers (GUIDs) and these terms are used interchangeably in the Microsoft protocol technical documents (TDs). Interchanging the usage of these terms does not imply or require a specific algorithm or mechanism to generate the UUID. Specifically, the use of this term does not imply or require that the algorithms described in [\[RFC4122\]](#) or [C706] has to be used for generating the UUID.

UTF-16LE: The Unicode Transformation Format - 16-bit, Little Endian encoding scheme. It is used to encode **Unicode** characters as a sequence of 16-bit codes, each encoded as two 8-bit bytes with the least-significant byte first.

Windows Internet Name Service (WINS): A name service for the NetBIOS protocol, particularly designed to ease transition to a TCP/IP based network. An implementation of an **NBNS server**.

WINS server: A server that hosts a Microsoft implementation of an **NBNS server**.

MAY, SHOULD, MUST, SHOULD NOT, MUST NOT: These terms (in all caps) are used as defined in [\[RFC2119\]](#). All statements of optional behavior use either MAY, SHOULD, or SHOULD NOT.

1.2 References

Links to a document in the Microsoft Open Specifications library point to the correct section in the most recently published version of the referenced document. However, because individual documents in the library are not updated at the same time, the section numbers in the documents may not match. You can confirm the correct section numbering by checking the [Errata](#).

1.2.1 Normative References

We conduct frequent surveys of the normative references to assure their continued availability. If you have any issue with finding a normative reference, please contact dochelp@microsoft.com. We will assist you in finding the relevant information.

[C706] The Open Group, "DCE 1.1: Remote Procedure Call", C706, August 1997, <https://publications.opengroup.org/c706>

Note Registration is required to download the document.

[ISO-8601] International Organization for Standardization, "Data Elements and Interchange Formats - Information Interchange - Representation of Dates and Times", ISO/IEC 8601:2004, December 2004, <http://www.iso.org/iso/en/CatalogueDetailPage.CatalogueDetail?CSNUMBER=40874&ICS1=1&ICS2=140&ICS3=30>

Note There is a charge to download the specification.

[MS-DTYP] Microsoft Corporation, "[Windows Data Types](#)".

[MS-ERREF] Microsoft Corporation, "[Windows Error Codes](#)".

[MS-NBTE] Microsoft Corporation, "[NetBIOS over TCP \(NBT\) Extensions](#)".

[MS-RPCE] Microsoft Corporation, "[Remote Procedure Call Protocol Extensions](#)".

[MS-WINSRA] Microsoft Corporation, "[Windows Internet Naming Service \(WINS\) Replication and Autodiscovery Protocol](#)".

[RFC1002] Network Working Group, "Protocol Standard for a NetBIOS Service on a TCP/UDP Transport: Detailed Specifications", STD 19, RFC 1002, March 1987, <https://www.rfc-editor.org/info/rfc1002>

[RFC2119] Bradner, S., "Key words for use in RFCs to Indicate Requirement Levels", BCP 14, RFC 2119, March 1997, <https://www.rfc-editor.org/info/rfc2119>

1.2.2 Informative References

[LMHOSTS] Microsoft Corporation, "LMHOSTS File Information and Predefined Keywords", February 2007, https://www.betaarchive.com/wiki/index.php?title=Microsoft_KB_Archive/102725

[MSDN-Handles] Microsoft Corporation, "Handles", [http://msdn.microsoft.com/en-us/library/aa373932\(VS.85\).aspx](http://msdn.microsoft.com/en-us/library/aa373932(VS.85).aspx)

[MSFT-ResourceKits] Microsoft Corporation, "Resource Kits", <http://technet.microsoft.com/en-us/library/cc875849.aspx>

1.3 Overview

The Remote Administrative Interface: WINS protocol is a client/server protocol that is used to remotely configure, manage, and monitor the **WINS server**. This protocol allows a **client** to view and

update the server configuration settings as well as to create, modify, and delete **WINS** database records. It also allows clients to trigger **scavenging** and replicating operations and to query the database.

The Remote Administrative Interface: WINS protocol is stateless with no state shared across **RPC** method calls. Each RPC method call contains one complete request. Output from one method call can be used as an input to another call, but the protocol does not provide methods for locking the WINS server configuration or state data across method calls. For example, a client can pull a range of records from the database and delete some of them using another RPC call. However, the protocol does not guarantee that the specified record has not been modified by another client between the two method calls.

Remote Administrative Interface: WINS	
Remote Procedure Call (RPC)	
TCP	Named Pipes

Figure 1: Relationship of Remote Administrative Interface: WINS to RPC

1.4 Relationship to Other Protocols

The Remote Administrative Interface: WINS protocol relies on **RPC** [\[MS-RPCE\]](#) as a transport. It is used to manage the **WINS** service on servers that implement the Windows Internet Naming Service (WINS) Replication and Autodiscovery Protocol [\[MS-WINSRA\]](#).

1.5 Prerequisites/Preconditions

The Remote Administrative Interface: WINS protocol is implemented on top of **RPC** and, as a result, has the prerequisites identified in [\[MS-RPCE\]](#).

The Remote Administrative Interface: WINS protocol assumes that before this protocol is invoked, a client has obtained the name or the IP address of the **WINS server** that implements this protocol suite.

1.6 Applicability Statement

The Remote Administrative Interface: WINS protocol is applicable when an application needs to remotely configure, manage, or monitor a **WINS server**.

Because the **NetBIOS** protocol [\[RFC1002\]](#) does not support the mapping between **NetBIOS names** and **IPv6** addresses, the Remote Administrative Interface: WINS protocol applies only to **IPv4** addresses. It does not apply to IPv6 addresses.

1.7 Versioning and Capability Negotiation

This document covers versioning issues in the following areas:

- **Supported Transports:** The Remote Administrative Interface: WINS protocol uses the **RPC** protocol as a transport and [RPC Protocol Sequences](#) as specified in section 2.1.
- **Protocol Versions:** This protocol has only one interface version, but that interface has been extended by adding additional methods at the end. The use of these methods is specified in section [3.1](#).
- **Security and Authentication Methods:** Authentication and security for the methods specified by this protocol are specified in [\[MS-RPCE\]](#) and in section 2.1.

- **Localization:** This protocol passes text strings in various methods. Localization considerations for such strings are specified in sections [2.2](#) and [3.1.4](#).
- **Capability Negotiation:** The Remote Administrative Interface: WINS protocol does not support interface version negotiation. Instead, this protocol uses the interface version number specified in the **interface definition language (IDL)** for versioning and capability negotiation.

1.8 Vendor-Extensible Fields

The Remote Administrative Interface: WINS protocol uses Win32 error codes as defined in [\[MS-ERREF\]](#) (section 2.2). Vendors SHOULD reuse those values with their indicated meanings. Choosing any other value runs the risk of a collision in the future.

1.9 Standards Assignments

Remote Administrative Interface: WINS protocol uses the following private assignments.

Parameter	Value	Reference
RPC interface universally unique identifier (UUID)	45f52c28-7f9f-101a-b52b-08002b2efabe	[C706] section A.2.5 winsif interface (section 3.1)
RPC interface UUID	811109bf-a4e1-11d1-ab54-00a0c91e9b45	[C706] section A.2.5 winsi2 interface (section 3.2)
Named pipe	"\\pipe\\WinsPipe"	Transport (section 2.1)

2 Messages

2.1 Transport

For the Remote Administrative Interface: WINS protocol, the **WINS** server MUST support the following **RPC** transports:

- RPC over **TCP**, with port selection performed dynamically by RPC.
- RPC over **named pipes**, with the **endpoint** name "\\pipe\\WinsPipe".

2.1.1 Server Security Settings

The Remote Administrative Interface: WINS protocol uses **security support provider (SSP)** security provided by **RPC** as specified in [\[MS-RPCE\]](#). The **WINS** RPC server uses the principal name "Wins" and the authentication service `RPC_C_AUTHN_WINNT`.

The WINS server MUST allow only authenticated access to RPC clients. The WINS server MUST NOT allow anonymous or unauthenticated RPC clients to connect. The WINS server MUST perform authorization checks to ensure that the client is authorized to perform a specific RPC operation.

The following mechanisms are enforced for client authorization:

- The WINSRA client SHOULD be a member of the WINS Users or WINS Administrator security group in order to retrieve information from the WINS server. This level of authorization is termed "query-level access".[<1>](#)
- The WINSRA client MUST be a member of the WINS Administrator security group before it can retrieve or modify information on the WINS server. This level of authorization is termed "control-level access".

Control-level access also includes query-level access. Therefore, clients with control access can also call methods that require only query-level access. The WINS server MUST limit access to only those clients that negotiate an **authentication level** equal to or higher than `RPC_C_AUTHN_LEVEL_CONNECT`.

2.1.2 Client Security Settings

The **RPC** client SHOULD use **security support provider (SSP)** security provided by RPC as specified in [\[MS-RPCE\]](#). The clients SHOULD use the server principal name "Wins", authentication service `RPC_C_AUTHN_WINNT`, and **authentication level** `RPC_C_AUTHN_LEVEL_CONNECT` while creating the binding handle.

2.2 Common Data Types

In addition to the **RPC** base types and definitions specified in [\[C706\]](#) and [\[MS-RPCE\]](#), additional data types are defined in this section. The following statements apply to those data types unless indicated otherwise:

- All multiple byte numeric values in messages use **little-endian** byte order.
- All character strings are encoded in **Unicode UTF-16LE**.

2.2.1 Datatypes, Enumerations, and Constants

2.2.1.1 WINSIF_HANDLE

The WINSIF_HANDLE data type is defined as a pointer to the [WINSINTF_BIND_DATA_T](#) structure. It is used by the **RPC** methods [R_WinsGetBrowserNames](#) and [R_WinsStatusWHdl](#).

This type is declared as follows:

```
typedef [handle] PWINSINTF_BIND_DATA_T WINSIF_HANDLE;
```

2.2.1.2 WINSINTF_VERS_NO_T

The WINSINTF_VERS_NO_T data type indicates the version number of a **WINS** database record. It is used by several **RPC** methods like [R_WinsGetDbRecs](#) and [R_WinsDelDbRecs](#).

This type is declared as follows:

```
typedef LARGE_INTEGER WINSINTF_VERS_NO_T;
```

2.2.1.3 WINSINTF_MAX_NO_RPL_PNRS

The WINSINTF_MAX_NO_RPL_PNRS constant defines the maximum number of **pull replication partners**. It is used by the structure [WINSINTF_RESULTS_T](#) (section 2.2.2.7).

Constant/value	Description
WINSINTF_MAX_NO_RPL_PNRS 25	The maximum number of pull replication partners.

2.2.1.4 WINSINTF_ACT_E

The WINSINTF_ACT_E enumeration indicates an action type requested by the **RPC** method [R_WinsRecordAction](#) for a record contained in the [WINSINTF_RECORD_ACTION_T](#) structure.

```
typedef enum _WINSINTF_ACT_E
{
    WINSINTF_E_INSERT = 0,
    WINSINTF_E_DELETE,
    WINSINTF_E_RELEASE,
    WINSINTF_E_MODIFY,
    WINSINTF_E_QUERY
} WINSINTF_ACT_E,
*PWINSINTF_ACT_E;
```

WINSINTF_E_INSERT: Insert a record into the **WINS** database.

WINSINTF_E_DELETE: Delete a matching record from the WINS database.

WINSINTF_E_RELEASE: Release a matching record from the WINS database.

WINSINTF_E_MODIFY: Modify the attributes of the matching record.

WINSINTF_E_QUERY: Query the database for a given name.

2.2.1.5 WINSINTF_CMD_E

The WINSINTF_CMD_E enumeration is used by the **RPC** methods to retrieve the configuration of a particular **WINS** server. This enumeration is used in conjunction with the [WINSINTF_RESULTS_T](#) and [WINSINTF_RESULTS_NEW_T](#) structures.

```
typedef enum WINSINTF_CMD_E
{
    WINSINTF_E_ADDVERSMAP = 0,
    WINSINTF_E_CONFIG,
    WINSINTF_E_STAT,
    WINSINTF_E_CONFIG_ALL_MAPS
} WINSINTF_CMD_E,
*PWINSINTF_CMD_E;
```

WINSINTF_E_ADDVERSMAP: Gets an entry from the **owner version map** of the **target WINS server**.

WINSINTF_E_CONFIG: Get the configuration details of the target WINS server.

WINSINTF_E_STAT: Get statistics for the target WINS server.

WINSINTF_E_CONFIG_ALL_MAPS: Get all owner version map entries from the target WINS server.

2.2.1.6 WINSINTF_TRIG_TYPE_E

The WINSINTF_TRIG_TYPE_E enumeration defines the type of **replication** to be done. It is used by the **RPC** method [R_WinsTrigger](#).

```
typedef enum _WINSINTF_TRIG_TYPE_E
{
    WINSINTF_E_PULL = 0,
    WINSINTF_E_PUSH,
    WINSINTF_E_PUSH_PROP
} WINSINTF_TRIG_TYPE_E,
*PWINSINTF_TRIG_TYPE_E;
```

WINSINTF_E_PULL: The **target WINS server** performs pull replication with the specified **WINS** server.

WINSINTF_E_PUSH: The target WINS server performs push replication with the specified WINS server.

WINSINTF_E_PUSH_PROP: The target WINS server performs propagating push replication with the specified WINS server.

2.2.1.7 WINSINTF_PRIORITY_CLASS_E

The WINSINTF_PRIORITY_CLASS_E enumeration defines the **priority class** of a **WINS** process. It is used by the **RPC** method [R_WinsSetPriorityClass](#).

```
typedef enum _WINSINTF_PRIORITY_CLASS_E
{
    WINSINTF_E_NORMAL = 0,
    WINSINTF_E_HIGH
}
```

```

} WINSINTF_PRIORITY_CLASS_E,
*PWINSINTF_PRIORITY_CLASS_E;

```

WINSINTF_E_NORMAL: WINS process is assigned normal priority class.

WINSINTF_E_HIGH: WINS process is assigned high priority class.

2.2.1.8 WINSINTF_SCV_OPC_E

The WINSINTF_SCV_OPC_E enumeration specifies the type of **scavenging** to be done on the **target WINS server**. This enumeration is used in the structure [WINSINTF_SCV_REQ_T](#).

```

typedef enum _WINSINTF_SCV_OPC_E
{
    WINSINTF_E_SCV_GENERAL = 0,
    WINSINTF_E_SCV_VERIFY
} WINSINTF_SCV_OPC_E,
*PWINSINTF_SCV_OPC_E;

```

WINSINTF_E_SCV_GENERAL: Requests normal scavenging operation.

WINSINTF_E_SCV_VERIFY: Verifies only the replicated **active records** with their **owner NBNS servers** for their validity.

2.2.2 Structures

2.2.2.1 WINSINTF_ADD_T

The WINSINTF_ADD_T structure defines the IP address information of a **WINS** server. It is used by several data structures including [WINSINTF_RECORD_ACTION_T](#) and [WINSINTF_ADD_VERS_MAP_T](#) and by RPC methods like [R_WinsTrigger](#) and [R_WinsGetDbRecs](#).

```

typedef struct _WINSINTF_ADD_T {
    BYTE Type;
    DWORD Len;
    DWORD IPAdd;
} WINSINTF_ADD_T,
*PWINSINTF_ADD_T;

```

Type: Specifies the address type. This field MUST be set to zero.

Len: Indicates the length, in bytes, of the IP address that is stored in **IPAdd**.

IPAdd: Stores an IP address in **little-endian** format. For example, the IP address 172.22.32.42 is stored as 0xAC16202A.

2.2.2.2 WINSINTF_BIND_DATA_T

The WINSINTF_BIND_DATA_T structure defines the binding information of the **WINS** server to which the client connects.

```

typedef struct _WINSINTF_BIND_DATA_T {
    DWORD fTcpIp;
    [string] LPSTR pServerAdd;
    [string] LPSTR pPipeName;
}

```



```

} WINSINTF_BIND_DATA_T,
*PWINSINTF_BIND_DATA_T;

```

fTcpIp: The transport mechanism to be used. If this value is 0x00000001, then TCP/IP is selected; otherwise, the **named pipe** is selected.

pServerAdd: A NULL-terminated string that specifies the server IP address.

pPipeName: A NULL-terminated string that specifies the pipe name. This value MUST be NULL when *fTcpIP* is 0x00000001.

2.2.2.3 WINSINTF_RECORD_ACTION_T

The WINSINTF_RECORD_ACTION_T structure defines a **WINS** database record and the action to be performed on it. The structure [WINSINTF_RECS_T \(section 2.2.2.8\)](#) and the **RPC** method [R_WinsRecordAction \(section 3.1.4.1\)](#) both use this structure.

```

typedef struct _WINSINTF_RECORD_ACTION_T {
    WINSINTF_ACT_E Cmd_e;
    [size_is(NameLen + 1)] LPSTR pName;
    DWORD NameLen;
    DWORD TypOfRec_e;
    DWORD NoOfAdds;
    [unique, size_is(NoOfAdds)] PWINSINTF_ADD_T pAdd;
    WINSINTF_ADD_T Add;
    LARGE_INTEGER VersNo;
    BYTE NodeType;
    DWORD OwnerId;
    DWORD State_e;
    DWORD fStatic;
    DWORD_PTR TimeStamp;
} WINSINTF_RECORD_ACTION_T,
*PWINSINTF_RECORD_ACTION_T;

```

Cmd_e: A [WINSINTF_ACT_E](#) enumeration (section 2.2.1.4) value that specifies the action to be performed on the specified record.

pName: A pointer to a NULL-terminated string that contains the **NetBIOS name** and optionally the **NetBIOS scope** name of the record. The NetBIOS scope name, if present, is appended to the NetBIOS name with a dot character ".".

If the NetBIOS name contains fewer than 16 characters, space characters MUST be used to pad the name string up to the **NetBIOS suffix**, which occupies the 16th character position.

NameLen: The length of the string that *pName* points to. It has the following possible values:

Value	Meaning
16	The pName value points to a string that contains only the NetBIOS name of the record. The NameLen value does not include the terminating NULL character.
18 < <i>value</i>	The pName value points to a string that contains the NetBIOS name, a dot character ".", and the NULL-terminated NetBIOS scope name of the record. The NameLen value includes the terminating NULL character. If the NameLen value is greater than 255, the pName string SHOULD be truncated to 254 characters plus a terminating NULL character.

TypOfRec_e: The record type. Only the two least-significant bits of the member value are considered valid. All other bits are masked with zero. The following values are allowed.

Value	Meaning
0	Unique name
1	Normal group name
2	Special group name
3	Multihomed machine name

NoOfAddrs: The number of IP addresses that are mapped to the NetBIOS name given in *pName*. It SHOULD have the value zero for unique names and **normal groups**, and it SHOULD have a value greater than 0x00000001 for other types of records.

pAdd: A pointer to an array of IP addresses that are mapped to the name given in *pName*. It MUST be used only for **multihomed (2)** and **special group** types of records.

Add: The IP address mapped to the name given in *pName*. This member MUST be used only for unique and normal group types of records.

VersNo: The version number of the record.

NodeType: The **NetBT node type**. Only the two least-significant bits of the member value are considered valid. All other bits are masked with zero. This member MUST have one of the following values:

Value	Meaning
0	B-node
1	P-node
2	M-node
3	H-node

OwnerId: The **owner** IP address of the record, in **little-endian** byte order.

State_e: The state of the record. Only the two least-significant bits of the member value are considered valid. All other bits are masked with zero. This member MUST have one of the following values:

Value	Meaning
0	Active record
1	Released record
2	Tombstoned record
3	Deleted record

fStatic: A value that indicates whether the record is static or dynamic. A value of 0 indicates a **dynamic record**, and 1 indicates a **static record**. Only the least-significant bit is considered valid. All other bits are masked with zero.

TimeStamp: The time stamp [\[ISO-8601\]](#) of the record.

2.2.2.4 WINSINTF_ADD_VERS_MAP_T

The WINSINTF_ADD_VERS_MAP_T structure defines an **address version map** pair. This data structure is generally used by other data structures, such as [WINSINTF_RESULTS_T](#) and [WINSINTF_RESULTS_NEW_T](#).

```
typedef struct _WINSINTF_ADD_VERS_MAP_T {
    WINSINTF_ADD_T Add;
    LARGE_INTEGER VersNo;
} WINSINTF_ADD_VERS_MAP_T,
*PWINSINTF_ADD_VERS_MAP_T;
```

Add: A structure containing the IP address of a partner **WINS server**.

VersNo: The highest version number from all of the records owned by a WINS server at the **target WINS server** database. Each record in the database has a version number and owner Id associated with it.

2.2.2.5 WINSINTF_RPL_COUNTERS_T

The WINSINTF_RPL_COUNTERS_T structure defines counters that contain the number of successful pull **replications** and the number of communication failures for a given **replication partner**. It is used in the structure [WINSINTF_STAT_T](#).

```
typedef struct _WINSINTF_RPL_COUNTERS_T {
    WINSINTF_ADD_T Add;
    DWORD NoOfRpls;
    DWORD NoOfCommFails;
} WINSINTF_RPL_COUNTERS_T,
*PWINSINTF_RPL_COUNTERS_T;
```

Add: The IP address of a **partner WINS** server.

NoOfRpls: The number of successful pull replications that have been performed with the replication partner. The **target WINS server** stores the replication partner's IP address in the *Add* member.

NoOfCommFails: The number of communication failures that have occurred in pull replications between the WINS server whose IP address is given in *Add* and the target WINS server.

2.2.2.6 WINSINTF_STAT_T

The WINSINTF_STAT_T structure defines counters, configured timestamps, the pull replication statistics for a given **WINS** server. This structure is used by the structure [WINSINTF_RESULTS_T](#) ([section 2.2.2.7](#)).

```
typedef struct _WINSINTF_STAT_T {
    struct {
        DWORD NoOfUniqueReg;
        DWORD NoOfGroupReg;
        DWORD NoOfQueries;
        DWORD NoOfSuccQueries;
        DWORD NoOfFailQueries;
        DWORD NoOfUniqueRef;
        DWORD NoOfGroupRef;
        DWORD NoOfRel;
        DWORD NoOfSuccRel;
        DWORD NoOfFailRel;
        DWORD NoOfUniqueCnf;
        DWORD NoOfGroupCnf;
    };
};
```

```

    } Counters;
    struct {
        SYSTEMTIME WINSStartTime;
        SYSTEMTIME LastPScvTime;
        SYSTEMTIME LastATScvTime;
        SYSTEMTIME LastTombScvTime;
        SYSTEMTIME LastVerifyScvTime;
        SYSTEMTIME LastPRplTime;
        SYSTEMTIME LastATRplTime;
        SYSTEMTIME LastNTRplTime;
        SYSTEMTIME LastACTRplTime;
        SYSTEMTIME LastInitDbTime;
        SYSTEMTIME CounterResetTime;
    } TimeStamps;
    DWORD NoOfPnrs;
    [unique, size_is(NoOfPnrs)] PWINSINTF_RPL_COUNTERS_T pRplPnrs;
} WINSINTF_STAT_T,
*PWINSINTF_STAT_T;

```

Counters: A structure that contains 32-bit unsigned integer counters, which measure various statistics on a WINS server.

NoOfUniqueReg: The number of unique registrations on the **target WINS server** since the service was started.

NoOfGroupReg: The number of group registrations at the target WINS server since the service was started.

NoOfQueries: The number of queries that clients have performed on the target WINS server to resolve NetBIOS names since the service was started. This value is the sum of the values maintained in *NoOfSuccQueries* and *NoOfFailQueries*.

NoOfSuccQueries: The number of successful **name resolution** queries on the target WINS server since the service was started.

NoOfFailQueries: The number of failed name resolution queries on the target WINS server since the service was started.

NoOfUniqueRef: The number of unique name refreshes on the target WINS server since the service was started.

NoOfGroupRef: The number of group name refreshes on the target WINS server since the service was started.

NoOfRel: The number of name releases on the target WINS server since the service was started. This value is the sum of the values maintained in *NoOfSuccRel* and *NoOfFailRel*.

NoOfSuccRel: The number of successful name releases on the target WINS server since the service was started.

NoOfFailRel: The number of failed name releases on the target WINS server since the service was started.

NoOfUniqueCnf: The number of unique name conflicts on the target WINS server since the service was started. Unique name conflicts can occur in the following cases:

- The server is registering or refreshing unique name requests from clients.
- The server is replicating unique name records from a partner WINS server.

NoOfGroupCnf: The number of group name conflicts on the target WINS server since the service was started. Group name conflicts can occur in the following cases:

- The server is registering or refreshing unique name requests from clients.
- The server is replicating unique name records from a partner WINS server.

TimeStamps: A structure that contains data in SYSTEMTIME structures ([\[MS-DTYP\]](#) section 2.3.13), which reflect the local time zone of the target WINS server.

WINSStartTime: The time at which the WINS service was started on the target WINS server.

LastPScvTime: The time at which the last periodic **scavenging** operation was done on the target WINS server.

LastATScvTime: The time at which the last administrator-triggered scavenging operation was done on the target WINS server.

LastTombScvTime: The time at which the last scavenging operation was done for the replicated tombstone records on the target WINS server.

LastVerifyScvTime: The time at which the last verification scavenging operation was done for the replicated active records on the target WINS server.

LastPRplTime: The time at which the last periodic pull replication was done on the target WINS server.

LastATRplTime: The time at which the last administrator-triggered pull replication was done on the target WINS server.

LastNTRplTime: This member is not set and MUST be ignored on receipt.

LastACTRplTime: This member is not set and MUST be ignored on receipt.

LastInitDbTime: The time at which the last static database initialization was done on the target WINS server.

CounterResetTime: The last time at which the administrator has cleared the success and failure replication counters of the target WINS server.

NoOfPnrs: The number of **pull partners** configured for the target WINS server.

pRplPnrs: A pointer to structures that contain the details of successful and failed **replication** counters of configured pull partners at the target WINS server, since the time service was started; or, the time at which the last reset happened by a call to the method [R_WinsResetCounters \(section 3.1.4.12\)](#). The number of structures is specified by **NoOfPnrs**.

2.2.2.7 WINSINTF_RESULTS_T

The WINSINTF_RESULTS_T structure defines information related to the configuration and statistics of a **target WINS server**. This is used by **RPC** method [R_WinsStatus](#).

```
typedef struct _WINSINTF_RESULTS_T {
    DWORD NoOfOwners;
    WINSINTF_ADD_VERS_MAP_T AddVersMaps[WINSINTF_MAX_NO_RPL_PNRS];
    LARGE_INTEGER MyMaxVersNo;
    DWORD RefreshInterval;
    DWORD TombstoneInterval;
    DWORD TombstoneTimeout;
    DWORD VerifyInterval;
    DWORD WINSPriorityClass;
    DWORD NoOfWorkers;
    WINSINTF_STAT_T WINSStat;
} WINSINTF_RESULTS_T;
```

`*PWINSINTF_RESULTS_T;`

NoOfOwners: The number of **owners** whose records are part of the target WINS server database. The value of this member MUST be less than or equal to 25.

AddVersMaps: A structure containing the **owner version map** of the target WINS server. The number of valid entries is defined by the *NoOfOwners* value.

MyMaxVersNo: This member is not set and MUST be ignored on receipt.

RefreshInterval: The refresh time interval configured on the target WINS server, in seconds.

TombstoneInterval: The **tombstone interval** configured on the target WINS server, in seconds.

TombstoneTimeout: The tombstone timeout configured on the target WINS server, in seconds.

VerifyInterval: The verify time interval configured on the target WINS server, in seconds.

WINSPriorityClass: The **priority class** of the **WINS** process running on the target WINS server. It SHOULD [<2>](#) have one of the following values:

Value	Meaning
NORMAL_PRIORITY_CLASS 0x00000020	The process has no special scheduling requirements.
HIGH_PRIORITY_CLASS 0x00000080	The process performs time-critical tasks that MUST be executed immediately for the process to run correctly. The threads of a high-priority class process preempt the threads of normal-priority class processes.

NoOfWorkersThds: The number of threads created in the WINS process for serving the **NetBIOS name** requests.

WINSStat: A [WINSINTF_STAT_T](#) structure (section 2.2.2.6) containing timing parameters configured on the target WINS server and the pull replication statistics of partner **WINS servers**.

2.2.2.8 WINSINTF_RECS_T

The structure `WINSINTF_RECS_T` defines an array of [WINSINTF_RECORD_ACTION_T](#) (section 2.2.2.3) elements. The [R_WinsGetDbRecs](#) (section 3.1.4.6) and [R_WinsGetDbRecsByName](#) (section 3.1.4.19) methods use this structure.

```
typedef struct _WINSINTF_RECS_T {  
    DWORD BuffSize;  
    [unique, size is (NoOfRecs)] PWINSINTF_RECORD_ACTION_T pRow;  
    DWORD NoOfRecs;  
    DWORD TotalNoOfRecs;  
} WINSINTF_RECS_T,  
*PWINSINTF_RECS_T;
```

BuffSize: The number of bytes allocated for the pointer *pRow*.

pRow: A pointer to an array of `WINSINTF_RECORD_ACTION_T` elements.

NoOfRecs: The number of records stored in the array pointed to by *pRow*.

TotalNoOfRecs: This member is not set and MUST be ignored on receipt.

2.2.2.9 WINSINTF_BROWSER_INFO_T

The WINSINTF_BROWSER_INFO_T structure defines information about browser names. It is used by the structure [WINSINTF_BROWSER_NAMES_T](#).

```
typedef struct WINSINTF_BROWSER_INFO_T {
    DWORD dwNameLen;
    [string] LPBYTE pName;
} WINSINTF_BROWSER_INFO_T,
*PWINSINTF_BROWSER_INFO_T;
```

dwNameLen: The length of the name that **pName** points to, in bytes. The length includes the terminating NULL character.

pName: A pointer to a NULL-terminated string that contains the **browser name**.

2.2.2.10 WINSINTF_BROWSER_NAMES_T

The WINSINTF_BROWSER_NAMES_T structure defines an array of browser names. This structure is used by the **RPC** method [R_WinsGetBrowserNames](#).

```
typedef struct WINSINTF_BROWSER_NAMES_T {
    DWORD EntriesRead;
    [unique, size is(EntriesRead)] PWINSINTF_BROWSER_INFO_T pInfo;
} WINSINTF_BROWSER_NAMES_T,
*PWINSINTF_BROWSER_NAMES_T;
```

EntriesRead: The number of entries in the array that **pInfo** points to.

pInfo: A pointer to an array of browser names. **EntriesRead** contains the length of this array.

2.2.2.11 WINSINTF_RESULTS_NEW_T

The WINSINTF_RESULTS_NEW_T structure defines configuration information and statistics for a **target WINS server**. This structure is used by the **RPC** method [R_WinsStatusNew \(section 3.1.4.20\)](#).

```
typedef struct _WINSINTF_RESULTS_NEW_T {
    DWORD NoOfOwners;
    [unique, size is(NoOfOwners)] PWINSINTF_ADD_VERS_MAP_T pAddVersMaps;
    LARGE_INTEGER MyMaxVersNo;
    DWORD RefreshInterval;
    DWORD TombstoneInterval;
    DWORD TombstoneTimeout;
    DWORD VerifyInterval;
    DWORD WINSPriorityClass;
    DWORD NoOfWorkersThds;
    WINSINTF_STAT_T WINSStat;
} WINSINTF_RESULTS_NEW_T,
*PWINSINTF_RESULTS_NEW_T;
```

NoOfOwners: The number of **owners** whose records are part of the target WINS server database.

pAddVersMaps: A pointer to an array of [WINSINTF_ADD_VERS_MAP_T](#) structure (section 2.2.2.4) elements. The **NoOfOwners** member contains the number of elements in the array.

MyMaxVersNo: This member is not set and MUST be ignored on receipt.

RefreshInterval: The refresh time interval configured on the target WINS server, in seconds.

TombstoneInterval: The **tombstone time interval** configured on the target WINS server, in seconds.

TombstoneTimeout: The tombstone timeout configured on the target WINS server, in seconds.

VerifyInterval: The verify time interval configured on the target WINS server, in seconds.

WINSPriorityClass: The priority class of the **WINS** process running on the target WINS server. It can have one of the following values:

Value	Meaning
NORMAL_PRIORITY_CLASS 0x00000020	The process has no special scheduling requirements.
HIGH_PRIORITY_CLASS 0x00000080	The process performs time-critical tasks that MUST be executed immediately for the process to run correctly. The threads of a high-priority class process preempt the threads of normal-priority class processes.

NoOfWorkerThds: The number of threads created in the WINS process to serve **NetBIOS name** requests.

WINSStat: A [WINSINTF_STAT_T](#) structure (section 2.2.2.6) containing timing parameters configured on the target WINS server and pull replication statistics of partner WINS servers.

2.2.2.12 WINSINTF_SCV_REQ_T

The WINSINTF_SCV_REQ_T structure defines the type of **scavenging** that needs to be done on the **target WINS server**. This is used by the **RPC** method [R_WinsDoScavengingNew](#) (section 3.1.4.22).

```
typedef struct _WINSINTF_SCV_REQ_T {  
    WINSINTF_SCV_OPC_E Opcode e;  
    DWORD Age;  
    DWORD fForce;  
} WINSINTF_SCV_REQ_T,  
*PWINSINTF_SCV_REQ_T;
```

Opcode_e: A [WINSINTF_SCV_OPC_E](#) enumeration (section 2.2.1.8) value describing the type of scavenging operation to be performed on the target **WINS** server.

Age: This member is not set and MUST be ignored on receipt.

fForce: Specifies whether a forceful scavenging is required.

Value	Meaning
0x00000000	The internal state and configuration of the WINS server determine whether scavenging is performed.
0x00000001 — 0xFFFFFFFF	The target WINS server performs scavenging.

3 Protocol Details

The client side of The Remote Administrative Interface: WINS protocol is simply a pass-through. This means that no additional timers or state are required on the client side of this protocol. Calls made by the higher-layer protocol or application are passed directly to the transport, and the results returned by the transport are passed directly back to the higher-layer protocol or application.

The **WINS** server supports two interfaces:

1. [winsif \(section 3.1.4\)](#)
2. [winsi2 \(section 3.2.4\)](#)

To support both interfaces, client applications are responsible for implementing mechanisms to manage memory such as the following:

- **midl_user_allocate**: Allocates memory for **RPC** input and output parameters.
- **midl_user_free**: Frees memory used by RPC input and output parameters.

Clients SHOULD call **midl_user_allocate** to allocate memory for any input pointer arguments to the RPC methods. The client RPC stub MUST send the input data to the server and then SHOULD free that memory by calling **midl_user_free**. Similarly, when the client RPC stub receives a response from server, it SHOULD call **midl_user_allocate** to allocate memory for all output pointer arguments. Client applications SHOULD free this memory by calling **midl_user_free**.

3.1 winsif Server Details

The methods supported by the **winsif** interface are specified in [Message Processing Events and Sequencing Rules \(section 3.1.4\)](#).

3.1.1 Abstract Data Model

This section describes a conceptual model that an implementation can maintain to participate in this protocol. The described organization is provided to facilitate the explanation of protocol behavior. This document does not mandate that implementations adhere to this model as long as their external behavior is consistent with that described in this document.

A **NetBIOS name server (NBNS)** maintains the following data structures:

Name record: A data structure that contains a name and the associated attributes.

Name records collection: A collection of all **name records** that are either registered by this NBNS server or obtained by **replication**.

Owner version map: A map between each NBNS **owner** and the record from that owner with the highest version number in the name records collection. This map determines whether the NBNS server pulls records from its partners and, if so, the range of records it obtains.

Global version counter: A 64-bit unsigned integer that tracks the version number given to the next record that is updated.

Server configuration: Parameters maintained in persistent storage include the following:

- Refresh interval
- **Extinction interval**
- Extinction timeout

- Verify interval
- Process **priority class**
- Number of worker threads

Browser name cache: A list of browser names that are stored in the **target WINS server**. When the **WINS** service is initialized, the cache SHOULD be empty.

The **Browser name cache** SHOULD be populated when the **R_WinsGetBrowserNames** method (section [3.1.4.18](#)) is called for the first time, and every subsequent call to that method SHOULD return the contents of the cache. If 3 minutes or more has elapsed from the time the **Browser name cache** is refreshed, the WINS service SHOULD get the name records from the target WINS server database and update the cache.

3.1.2 Timers

No timers are required beyond those used internally by **RPC** to implement resiliency to network outages, as specified in [\[MS-RPCE\]](#) section 3.2.3.2.1.

3.1.3 Initialization

The **winsif** server for the Remote Administrative Interface: WINS MUST be initialized by registering the **RPC** interface and listening on the dynamically allocated port assigned by RPC, as specified in section [2.1](#). The client MUST contact the well-known RPC port on the **WINS** server to find the **endpoint** of **winsif**.

3.1.4 Message Processing Events and Sequencing Rules

The winsif interface provides methods that remotely configure, manage and monitor the **WINS** server.

Methods in RPC Opnum Order

Method	Description
R_WinsRecordAction	Inserts, modifies, deletes, releases or queries a Name Record from the WINS database. Opnum: 0
R_WinsStatus	Retrieves various counters, configuration settings and the statistics of a WINS server. Opnum: 1
R_WinsTrigger	Queues a request to trigger a replication from target WINS server to a specified WINS server. Opnum: 2
R_WinsDoStaticInit	Imports Name Records from a file to the WINS database. Opnum: 3
R_WinsDoScavenging	Queues a Scavenging request at the target WINS server. Opnum: 4
R_WinsGetDbRecs	Retrieves Name Records lying in a range of two version numbers and are owned by a particular WINS server. Opnum: 5
R_WinsTerm	Sends a termination signal to the WINS process running on the target WINS

Method	Description
	server. Opnum: 6
R_WinsBackup	Backs up the WINS database to a specified directory. Opnum: 7
R_WinsDelDbRecs	Deletes Name Records lying in a range of two version numbers and are owned by a particular WINS server. Opnum: 8
R_WinsPullRange	Pulls a range of records owned by a particular WINS server from another WINS server and replicates them with the target WINS server database. Opnum: 9
R_WinsSetPriorityClass	Modifies the priority class of a WINS process running on the target WINS server. Opnum: 10
R_WinsResetCounters	Resets all the pull replication partners counters stored at the target WINS server. Opnum: 11
R_WinsWorkerThdUpd	Modifies the number of NetBIOS threads to a new value at the target WINS server. Opnum: 12
R_WinsGetNameAndAdd	Retrieves the NetBIOS name and the corresponding IP address of the target WINS server. Opnum: 13
R_WinsGetBrowserNames_Old	This method SHOULD not be used. Opnum: 14
R_WinsDeleteWins	Deletes all the records owned by a particular WINS server from the target WINS server database. Opnum: 15
R_WinsSetFlags	This method SHOULD not be used. Opnum: 16
R_WinsGetBrowserNames	Retrieves the Browser Names information stored at the target WINS server. Opnum: 17
R_WinsGetDbRecsByName	Retrieves records matching a specified owner address from the target WINS server database. Opnum: 18
R_WinsStatusNew	Retrieves various configuration settings and the statistics of a WINS server. Opnum: 19
R_WinsStatusWHdl	Retrieves various configuration settings and the statistics of a WINS server. Opnum: 20
R_WinsDoScavengingNew	Requests a Scavenging operation at the target WINS server. Opnum: 21

3.1.4.1 R_WinsRecordAction (Opnum 0)

The R_WinsRecordAction method inserts, modifies, deletes, releases, or queries a **name record** from the **WINS** database.

```
DWORD R_WinsRecordAction(  
    [in] handle_t ServerHdl,  
    [in, out, ref] PWINSINTF_RECORD_ACTION_T* ppRecAction  
);
```

ServerHdl: An **RPC** binding over IP address/**HostName** to the **WINS server**. RPC uses this binding internally to determine which WINS server the call is directed to. <3>

ppRecAction: A pointer to a [WINSINTF_RECORD_ACTION_T](#) structure (section 2.2.2.3) that contains the details of the record and the action to be performed on it. The interpretation of the member values in this structure depends on the type of action specified by the [WINSINTF_ACT_E](#) enumeration (section 2.2.1.4) value in its **Cmd_e** member, as follows.

WINSINTF_E_INSERT:

- **Cmd_e** is set to WINSINTF_E_INSERT.
- **pName** points to a NULL-terminated string that contains the **NetBIOS name** and optionally the **NetBIOS scope** name of the record.
- **NameLen** contains the length of the string specified by **pName**.
- **TypOfRec_e** is set to a value between 0x00000000 and 0x00000003 based on the record type.
- **NoOfAdds** is set to a positive value based on the number of IP address mappings that the record has.
- **pAdd** or **Add** is set with the mapping IP addresses, based on the **TypOfRec_e** member.
- **VersNo** SHOULD be ignored by the server. The inserted record MUST be marked with the current version number that is in use at the WINS server.
- **NodeType** is set to a value between 0x00 and 0x03 based on the type of the node.
- **OwnerId** SHOULD be ignored by the server. The record MUST be inserted into the database with the **OwnerId** member set to the **target WINS server** address.
- **State_e** SHOULD be ignored by the server. The record MUST be inserted into the database with its state marked as **ACTIVE**.
- **fStatic** is set to 0x00000001 if the record being inserted is a static record; otherwise, it is set to 0x00000000.
- **TimeStamp** SHOULD be ignored by the server. The inserted record SHOULD be time-stamped with zero if the **fStatic** member is set to 0x00000001; otherwise, it SHOULD be time-stamped with the current time on the server plus the refresh interval configured on the server.

WINSINTF_E_DELETE:

- **Cmd_e** is set to WINSINTF_E_DELETE.
- **pName** points to a NULL-terminated string that contains the NetBIOS name and optionally the NetBIOS scope name of the record to be deleted from the database.

- **NameLen** contains the length of the string specified by **pName**.
- **State_e** is set to 0x00000003.
- All other members SHOULD be ignored by the server.
- WINSINTF_E_RELEASE:
- **Cmd_e** is set to WINSINTF_E_RELEASE.
- **pName** points to a NULL-terminated string that contains the NetBIOS name and optionally the NetBIOS scope name of the record.
- **NameLen** contains the length of the string specified by **pName**.
- **TypOfRec_e** is set to a value between zero and 0x00000003 based on the record type.
- **NoOfAdds** MUST be set to 0x00000001.
- **pAdd** or **Add** is set with the mapping IP address based on the **TypOfRec_e** member.
- **VersNo**, **NodeType**, **OwnerId**, and **fStatic** SHOULD be ignored by the server.
- **State_e** SHOULD be ignored by the server. The record MUST be inserted with state marked as **RELEASED**.
- **TimeStamp** SHOULD be ignored by the server. The released record SHOULD be time-stamped with 0xFFFFFFFF if the **fStatic** member is set to 0x00000001; otherwise, it SHOULD be time-stamped with the current time on the server plus the **tombstone interval** configured on the server.

WINSINTF_E_MODIFY:

- **Cmd_e** is set to WINSINTF_E_MODIFY.
- **pName** points to a NULL-terminated string that contains the NetBIOS name and optionally the NetBIOS scope name of the record to be modified in the database.
- **NameLen** contains the length of the string specified by **pName**.
- **TypOfRec_e** contains the record type to be set for the record matching the **pName** member in the WINS database.
- **NodeType** contains the node type to be set for the record matching the **pName** member in the WINS database.
- **State_e** contains the state to be set for the record matching the **pName** member in the WINS database.
- **fStatic** contains the value to be set for the record matching the **pName** member in the WINS database.
- All other members SHOULD be ignored by the server.

WINSINTF_E_QUERY:

- **Cmd_e** is set to WINSINTF_E_QUERY.
- **pName** points to a NULL-terminated string that contains the NetBIOS name and optionally the NetBIOS scope name of the record to be queried from the database.
- **NameLen** contains the length of the string specified by **pName**.

- All other members act as output, which are filled by the server if a matching entry is found in the database.
- **TypOfRec_e** contains the matching record type.
- If the **TyeOfRec_e** member is set to 0x00000000 or 0x00000001, the **NoOfAdds** member SHOULD contain 0x00000001 or the number of IP addresses that are mapped to the name given in the **pName** member.
- If the **TypOfRec_e** member is set to 0x00000002 or 0x00000003. The RPC method caller SHOULD refer to this member for the set of IP addresses mapped to the name given in the **pName** member.
- If the **TypOfRec_e** member is set to 0x00000000 or 0x00000001. The RPC method caller SHOULD refer to this member for the IP address mapped to the name given in the **pName** member. If the **TypOfRec_e** member is set to 0x00000001, the **IPAdd** member of the **Add** structure MUST contain 0xFFFFFFFF.
- **VersNo** contains the version number of the matching record.
- **NodeType** contains the node type of the matching record.
- **OwnerId** contains the IP address of the **owner** of the matching record.
- **State_e** contains the state of the matching record.
- **fStatic** contains the value 0x00000001 if the record is entered into the database by an administrator; otherwise, it contains 0x00000000.
- **TimeStamp** contains the time stamp of the record.

Return Values: A 32-bit unsigned integer value that indicates return status. A return value of ERROR_SUCCESS indicates that the operation was completed successfully. Otherwise, the **TimeStamp** member SHOULD contain one of the following Win32 error codes, as specified in [\[MS-ERREF\]](#):

Return value/code	Description
0x00000000 ERROR_SUCCESS	The call was successful.
0x00000FA0 ERROR_WINS_INTERNAL	An error occurred while processing the RPC call.
0x00000FA5 ERROR_REC_NON_EXISTENT	The name does not exist in the database. This error is returned only if a requested WINSINTF_E_QUERY operation is not successful.
0x00000005 ERROR_ACCESS_DENIED	The caller doesn't have sufficient permissions.

Exceptions Thrown: No exceptions are thrown beyond those thrown by the underlying RPC protocol [\[MS-RPCE\]](#).

Processing and Response Requirements:

The **Opnum** value for this method is 0.

When processing this call, the WINS server MUST do the following:

- If the action specified is WINSINTF_E_QUERY, the RPC method caller SHOULD have query level access. [<4>](#) For all other actions the caller SHOULD have control level access. If an RPC client with a lower access level calls this method, the server SHOULD return ERROR_ACCESS_DENIED.
- The WINS service on the target WINS server MUST be in the running or paused state for this method to complete successfully. If the service is in initializing or exiting state, ERROR_WINS_INTERNAL status SHOULD be returned.
- When the RPC method is called with an action set to WINSINTF_E_INSERT, the requested record is inserted into the database. If the record with the same name already exists in the database, name resolution occurs as described in [\[MS-WINSRA\]](#). The server returns ERROR_WINS_INTERNAL, if any error occurs while performing the resolution or inserting the record. [<5>](#)
- When an RPC method is called with the action set to WINSINTF_E_RELEASE, the state of the matching record is changed to RELEASED in the database. If a matching record is not found, the server returns ERROR_SUCCESS. If any failure occurs during the modification of the record state, ERROR_WINS_INTERNAL is returned.
- When an RPC method is called with the action set to WINSINTF_E_MODIFY, the database is searched for a matching record. If a match is found, the attributes of the record such as record type, node type, record state, and *fstatic* are modified according to the requested values. If the matching record's type is either unique or **Normal Group** and a request comes to modify it to multihomed or **Special Group**, respectively, an ERROR_WINS_INTERNAL error is returned; otherwise, ERROR_SUCCESS is returned. If the record is not found in the database, the server returns ERROR_SUCCESS.
- When the RPC method is called with the action set to WINSINTF_E_QUERY, the database is queried for the given name. If a matching record is found, the attributes of the record are returned to the RPC caller. If the record is not found or if any error occurs during attribute retrieval, the server returns an ERROR_REC_NON_EXISTENT error.
- When the RPC method is called with the action set to WINSINTF_E_DELETE, the matching record is deleted from the database. If a matching record is not found in the database, an ERROR_SUCCESS status code is returned. If any error occurs during the database operations, an ERROR_WINS_INTERNAL is returned. The RPC method caller MUST set *state_e* to DELETED for this action to succeed.

3.1.4.2 R_WinsStatus (Opnum 1)

The R_WinsStatus method retrieves configuration settings and statistics from a **WINS** server.

```
DWORD R_WinsStatus(
    [in] handle_t ServerHdl,
    [in] WINSINTF_CMD_E Cmd_e,
    [in, out, ref] PWINSINTF_RESULTS_T pResults
);
```

ServerHdl: An **RPC** binding over IP address/HostName to the WINS server. RPC uses this binding internally to determine which WINS server the call is directed to.

Cmd_e: The command to be executed on the **target WINS server** from the [WINSINTF_CMD_E](#) enumeration (section 2.2.1.5).

pResults: A pointer to a [WINSINTF_RESULTS_T](#) structure (section 2.2.2.7) that contains configuration data and statistics for the target WINS server.

Return Values: A 32-bit unsigned integer value that indicates return status. A return value of ERROR_SUCCESS (0x00000000) indicates that the operation completed successfully. Otherwise,

this return value is a Win32 error code, as specified in [\[MS-ERREF\]](#). The following Win32 error codes can be returned.

Return value/code	Description
0x00000000 ERROR_SUCCESS	The call was successful.
0x00000FA0 ERROR_WINS_INTERNAL	An error occurred while processing the RPC call.
0x00000005 ERROR_ACCESS_DENIED	The caller doesn't have sufficient permissions.

Exceptions Thrown: No exceptions are thrown beyond those thrown by the underlying RPC protocol [\[MS-RPCE\]](#).

Processing and Response Requirements:

The following requirements and recommendations apply to the WINS server that processes a call to R_WinStatus:

- The R_WinStatus caller SHOULD have query level access. [<6>](#) If an RPC client with a lower access level calls R_WinStatus, the server SHOULD return ERROR_ACCESS_DENIED.
- The WINS service on the target WINS server MUST be in the running or paused state. If the service is in initializing or exiting state, the server SHOULD return ERROR_WINS_INTERNAL.
- When R_WinStatus is called with the **Cmd_e** parameter set to WINSINTF_E_ADDVERSMAP, the first entry of the *AddVersMaps* array SHOULD contain the address of the **WINS server** for which the version number is requested. The WINSINTF_E_ADDVERSMAP command retrieves the version number for the specified **owner** address from the **owner version map** of the target WINS server. The retrieved version number is stored in the *AddVersMaps[0].VersNo* parameter. If the address of the **owner WINS server** is not found in the owner-version map of the target WINS server, an ERROR_WINS_INTERNAL error is returned.
- When R_WinStatus is called with the *Cmd_e* parameter set to WINSINTF_E_CONFIG, the *pResults* parameter is used only for output. The *NoOfOwners* and *AddVersMaps* parameters specify the owner version map table maintained on the target WINS server. If the owner version map table has more than 25 entries, only the first 25 entries are copied to *pResults->AddVersMaps*. The *RefreshInterval*, *TombstoneInterval*, *TombstoneTimeout*, *VerifyInterval*, *WINSPriorityClass*, and *NoOfWorkerThds* members get values according to the configuration of the target WINS server. The *WINSStat* parameter is not used for this command. An ERROR_WINS_INTERNAL error is returned if any error occurs during processing of a WINSINTF_E_CONFIG command.
- When R_WinStatus is called with the **Cmd_e** parameter set to WINSINTF_E_CONFIG_ALL_MAPS, the behavior is the same as specified for the WINSINTF_E_CONFIG command except that the owner version map entry is returned even if it is marked as deleted.
- When R_WinStatus is called with the **Cmd_e** parameter set to WINSINTF_E_STAT, the *pResults* parameter is used only for output. Statistics for the target WINS server are copied to *pResults->WINSStat*. The *pResults->WINSStat.pRplPnrs* pointer MUST be NULL for this operation to succeed. The WINSINTF_E_STAT command also retrieves the information retrieved by the WINSINTF_E_CONFIG command. An ERROR_WINS_INTERNAL error is returned if any error occurs during processing of a WINSINTF_E_STAT command.

3.1.4.3 R_WinsTrigger (Opnum 2)

The R_WinsTrigger method triggers a replication operation between a **target WINS server** and another **WINS** server.

```
DWORD R WinsTrigger(  
    [in] handle_t ServerHdl,  
    [in, ref] PWINSINTF ADD T pWinsAdd,  
    [in] WINSINTF_TRIG_TYPE_E TrigType_e  
);
```

ServerHdl: An **RPC** binding over IP address/HostName to the WINS server. RPC uses this binding internally to determine which WINS server the call is directed to.

pWinsAdd: Address of the WINS server with which the target WINS server performs the replication operation.

TrigType_e: The type of replication operation requested.

Return Values: A 32 bit unsigned integer value that indicates the return status. A return value of ERROR_SUCCESS (0x00000000) indicates that the operation completed successfully. Any other return value is a Win32 error code, as specified in [\[MS-ERREF\]](#). The following Win32 error codes can be returned:

Return value/code	Description
0x00000000 ERROR_SUCCESS	The call was successful.
0x00000FA0 ERROR_WINS_INTERNAL	An error occurred while processing the RPC call.
0x00000FA6 ERROR_RPL_NOT_ALLOWED	The WINS server requested for the replication operation is requested is not configured as a replication partner for the target WINS server.
0x00000005 ERROR_ACCESS_DENIED	The caller does not have sufficient permissions.

Exceptions Thrown: No exceptions are thrown beyond those thrown by the underlying RPC protocol [\[MS-RPCE\]](#).

Processing and Response Requirements:

When R_WinsTrigger is called, the server returns immediately without waiting for the replication operation to finish. The server just queues a request for the replication operation, and the replication takes place at a time determined by the internal state and configuration of the target WINS server. Hence, applications that call R_WinsTrigger SHOULD NOT treat an ERROR_SUCCESS return value as indicating a successful replication operation. Instead, applications SHOULD rely on WINS event logs to determine whether or not replication is successful. The following table lists the events that indicate the status of replication.

Event ID	Event Name	Event Description
Informational events		
4102	WINS_EVT_CONN_ABORTED	The connection was aborted by the remote WINS. It is possible that Remote WINS is not configured

Event ID	Event Name	Event Description
		to replicate with the server.
4108	WINS_EVT_CANT_GET_INITRPL_VAL	WINS could not read the InitTimeReplication field of the pull/push key.
4115	WINS_EVT_CANT_GET_CC_MAX_RECS_AAT_VAL	WINS could not read the MaxRecsAtATime value (type DWORD) in the Wins\Parameters\ConsistencyCheck subkey of the registry. Set this value so that WINS does not replicate more than a set number of records in one cycle while doing periodic consistency checks of the WINS database. When doing a consistency check, WINS replicates all records of an owner WINS by either going to that WINS or to a replication partner. At the end of a consistency check for an owner's records, WINS checks to see if it has replicated more than the specified values in the current consistency check cycle. If the value has been exceeded, the consistency check stops; otherwise it continues. In the next cycle, it starts from where it left off and returns to the first owner if required.
4116	WINS_EVT_CANT_GET_CC_USE_RPL_PNRS_VAL	WINS could not read the UseRplPnrs value of the Wins\Parameters\ConsistencyCheck key. If this value is set to a nonzero (DWORD) value, WINS will do a consistency check of the owners in its database by going to one or more of its replication partners. If the owner of the records happens to be a replication partner, WINS will go to it; otherwise it will pick a partner at random. Set this value if there is a large number of WINSs in the configuration and/or if the local WINS SHOULD NOT to go to any WINS that is not a replication partner.
4121	WINS_EVT_NO_RPL_RECS_RETRIEVED	The WINS Replicator could not find any records in the WINS database. This means there are no active or tombstone records in the database. It could be that the records being requested by a remote WINS server have either been released or do not exist.
4124	WINS_EVT_UPD_NTF_NOT_ACCEPTED	The WINS server received an update notification from the non-configured WINS server at the address, %1. The WINS server rejected it. This means the remote WINS server is not in the list of Push partners (WINS servers under the Pull key) and the administrator has prohibited (by using the registry) replication with non-configured WINS servers. To have this WINS server to accept update notifications from non-configured WINS servers then set the Wins\Parameters\RplOnlyWCnfPnrs value in the registry to 0.
4126	WINS_EVT_ADD_VERS_MAP_REQ_NOT_ACCEPTED	The WINS server received a pull request from the non-configured WINS server with the address, %1. The WINS server rejected it since the remote WINS server is not in the list of Pull partners (WINS servers under the Pull key) and the administrator has prohibited (by using the

Event ID	Event Name	Event Description
		registry) replication with non-configured partners. To have this WINS server to accept update notifications from WINS servers not in the "pull partner" list, then set the "replication only with configured partners" value in the registry to 0.
4134	WINS_EVT_INF_REM_WINS	The local WINS is informing a remote WINS to update the version number of a record. This is because the local WINS encountered a conflict between an active owned name and an active replica that it pulled from a replication partner.
4135	WINS_EVT_REM_WINS_INF	The local WINS has been informed by a remote WINS with the address %1, to update the version number of a record. This is because the remote WINS encountered a conflict between an active owned name and an active replica that it pulled from a replication partner.
4137	WINS_EVT_NAME_MISMATCH	A name mismatch was reported while verifying the validity of old replicas. The local record has the name %1 while the replica pulled in from the WINS that owns this record has the name %2. This could mean that the remote WINS was stopped and then restarted but its version counter value was not set to its previous value before termination.
4139	WINS_EVT_CNF_CHANGE	The WINS replication request is being ignored because WINS found that the Wins\Partners key information has changed (due to a notification from the registry) which makes the current partner request out-of-date.
4227	WINS_EVT_CANT_FIND_ANY_REC_IN_RANGE	The Push Thread was requested for a range of records but could not find any records in the range. The replication time intervals MUST be set properly. If the tombstone interval and timeout intervals are not correct (that is, much less than the replication interval), the preceding condition is possible. This is because the records might get changed into tombstones and then deleted before the remote WINS can pull them. Similarly, if the refresh interval is set to be much less than the replication interval, then the records could get released before a WINS can pull them (a released record is not sent).
4251	WINS_EVT_CONN_RETRIES_FAILED	The WINS Replication Pull Handler could not connect to a WINS server. All retries failed. WINS will try again after a set number of replication time intervals have elapsed.
4260	WINS_EVT_RPL_REG_ERR	WINS received an error while registering replicas. It will not register any additional replicas of this WINS at this time (the address is in the data section fourth through eighth byte). A previous log entry will specify the reason for this. If the same error occurs during subsequent replication with the preceding partner WINS, restore the WINS database from the backup.

Event ID	Event Name	Event Description
4261	WINS_EVT_RPL_REG_GRP_MEM_ERR	WINS received an error while trying to register a group's replica with name %1. The replica is owned by the WINS with the address given in the data section.
4262	WINS_EVT_RPL_REG_UNIQUE_ERR	WINS received an error while trying to register a unique replica with name %1. The replica is owned by WINS with the address in the data section.
4268	WINS_EVT_RPL_STATE_ERR	WINS received a replica whose state is incorrect. For example, the state might be RELEASED or the replica might be an Internet group that does not have any members but the state is not TOMBSTONE.
4289	WINS_EVT_RECORD_NOT_OWNED	WINS is trying to update the version number of a database record that it does not own. This is a serious error if the WINS server is updating the record after a conflict. It is not a serious error if the WINS server is updating the record as a result of a request to do so from a remote WINS server. When a remote WINS server notices a conflict between an active owned entry and a replica, it requests the replica owner to update the version number of the record. It is possible that the replica is no longer owned by the remote WINS.
4295	WINS_EVT_ADJ_VERS_NO	When WINS replicated with its partners, one of the partners showed there was more data that actually existed. WINS adjusted its counter so that new registrations and updates are seen by its partners. This means that recovery did not work properly. Check which of the partners has the highest version number corresponding to the local WINS. This can be accomplished by shutting down WINS and restarting after specifying this number in the registry.
4307	WINS_EVT_RPLPULL_EXC	The WINS replicator Pull thread encountered an error while processing a request. Log entries will specify what went wrong.
4312	WINS_EVT_TERM_DUE_TIME_LMT	WINS has exceeded the wait time for all threads to terminate. The number of threads still active is given in the second DWORD of the data section. The thread that could be stuck is the replicator thread, which could be stuck because the other WINS is slow in sending data or reading data. The latter can contribute to pressure on the TCP connection on which it is trying to replicate.
Warning events		
4153	WINS_EVT_UNABLE_TO_VERIFY	The Scavenger thread found active replicas that are required to be verified with the owner WINS server because they were older than the verify time interval. The table of owner-to-address mappings indicated the WINS was not active.
4155	WINS_EVT_REPLICA_CLASH_W_STATIC	A replica clashed with the static record, %1, in the WINS database. The replica was rejected.

Event ID	Event Name	Event Description
4161	WINS_EVT_PARTIAL_RPL_TYPE	A nonzero replication type applies for this partner, which means only a subset of records will be replicated between the local WINS and this partner. To get records that did not replicate, either pull them by using the winscl.exe in the Windows 2000 Resource Kit, (as described in [MSFT-ResourceKits]) or delete all owners acquired only through this partner and initiate replication after that to reacquire all their records. The partner's address is given in the second DWORD of the data section.
4162	WINS_EVT_PNR_PARTIAL_RPL_TYPE	A partner has requested only a subset of records. This means that all the records in the range requested will not be replicated. Check the partner's registry to see what replication type applies to it. The partner's address is given in the second DWORD of the data section.
4163	WINS_EVT_ADJ_MAX_RECS_AAT	WINS adjusted the Maximum Records at a time parameter of the ConsistencyCheck key. The value specified, %2, was changed to the minimum value, %1. This represents the maximum number of records that will be replicated at any one time for a consistency check.
4164	WINS_EVT_FORCE_SCV_R_T	WINS was forced to scavenge replica tombstones of a WINS. The administrator forced the scavenging by using winscl.exe. WINS does not scavenge replica tombstones unless they have timed out and the WINS has been running for at least three days. This is to ensure that the tombstones have replicated to other WINSes. In this case, the tombstones were timed out but the WINS had not been running for three days. The replica tombstones were deleted. This deletion does not constitute a problem unless there are WINS servers that are primary and backup to clients but not both Push and Pull partners of each other. If this type of WINS exists, there is a low probability that this action will result in database inconsistency but if it does, a consistent state can be achieved by initiating consistency checks by using winscl.exe.
Error events		
4166	WINS_EVT_RPLPULL_ABNORMAL_SHUTDOWN	The replication Pull thread is shutting down due to an error. Restart WINS.
4167	WINS_EVT_RPLPUSH_ABNORMAL_SHUTDOWN	The replication Push thread is shutting down due to an error. Restart WINS.
4197	WINS_EVT_WINSOCK_BIND_ERR	An address could not bind to a socket. Make sure the TCP/IP stack is installed and running properly. This event might mean that the 'nameserver' port (specified in the services file) which is used as the default by WINS for replication and discovering other WINSes has been taken by another process or service running on this computer. There are two options: either end the other process or service, or direct WINS to use another port. If you choose the

Event ID	Event Name	Event Description
		second option, set the value 'PortNo' (REG_DWORD) under the Wins\Parameters subkey in the registry to 1512. Note Changing the port number this way will prevent this WINS from replicating or discovering other WINs unless they too are directed to use the same port number as this WINS.

The following requirements and recommendations apply to a WINS server that processes a call to R_WinsTrigger:

- The R_WinsTrigger caller SHOULD have control level access. If an RPC client with a lower access level calls R_WinsTrigger, the server SHOULD return ERROR_ACCESS_DENIED.
- The WINS service on the target WINS server MUST be in the running or paused state. If the service is in initializing or exiting state, the server SHOULD return ERROR_WINS_INTERNAL for its status.
- When R_WinsTrigger is called with trigger type WINSINTF_E_PUSH, the server queues a push replication. If the target WINS server is configured to replicate only with partners and the address of the requested replication partner is not in the server's list of push replication partners, the server SHOULD return WINSINTF_RPL_NOT_ALLOWED. The server can also return ERROR_WINS_INTERNAL for any other errors occur while it processes the request.
- The trigger type WINSINTF_E_PUSH_PROP works same way as the command WINSINTF_E_PUSH except that the update notifications that are sent as part of push replication have the propagate opcode set (see [\[MS-WINSRA\]](#) section 2.2.8.
- When the R_WinsTrigger method is called with trigger type WINSINTF_E_PULL, the server queues a pull replication as specified by the *pWinsAdd* parameter. If the target WINS server is configured to replicate only with partners, and the address of the requested replication partner is not in the server's list of pull replication partners, the server SHOULD return WINSINTF_RPL_NOT_ALLOWED. Also, the server SHOULD return ERROR_WINS_INTERNAL for any other errors that occur while it processes the request.

3.1.4.4 R_WinsDoStaticInit (Opnum 3)

The R_WinsDoStaticInit method performs static initialization of a **WINS** database by registering the names specified in a data file.

```

DWORD R_WinsDoStaticInit(
    [in] handle t ServerHdl,
    [in, unique, string] LPWSTR pDataFilePath,
    [in] DWORD fDel
);

```

ServerHdl: An **RPC** binding over IP address/HostName to the WINS server. RPC uses this binding internally to resolve which WINS server the call is directed to.

pDataFilePath: A pointer to a **Unicode string** containing the path to a text file on the **target WINS server**. The file SHOULD contain entries that map **NetBIOS** names to **IPv4 addresses in string format** using the following syntax:

```
<IPv4 address 1> <one or more spaces> <NetBIOS name 1>
```

```
<IPv4 address 2> <one or more spaces> <NetBIOS name 2>
...
<IPv4 address N> <one or more spaces> <NetBIOS name N>
```

An example of this syntax can be found in the Windows **LMHOSTS** file, as specified in [\[MS-NBTE\]](#) section 2.2.3 LMHOSTS File Syntax. See [\[LMHOSTS\]](#) for more information.

If this pointer value is NULL, the target WINS server SHOULD use the following default path:
"%systemroot%\system32\drivers\etc\lmhosts".

fDel: Value specifying whether or not to delete the file specified by *pDataFilePath* from the target WINS server. A non-zero value deletes the file from the target WINS server after the database initialization is complete.

Return Values: A 32 bit unsigned integer value that indicates the return status. A return value of ERROR_SUCCESS (0x00000000) indicates that the operation completed successfully. Any other return value is a Win32 error code as specified in [\[MS-ERREF\]](#). The following Win32 error codes can be returned:

Return value/code	Description
0x00000000 ERROR_SUCCESS	The call was successful.
0x00000FA2 ERROR_STATIC_INIT_FAILED	An error occurred during static initialization of the database file.
0x00000005 ERROR_ACCESS_DENIED	The caller does not have sufficient permissions.

Exceptions Thrown: No exceptions are thrown beyond those thrown by the underlying RPC protocol [\[MS-RPCE\]](#).

Processing and Response Requirements:

The following requirements and recommendations apply to a WINS server that processes a call to R_WinsDoStaticInit:

- The R_WinsDoStaticInit caller SHOULD have control level access. If an RPC client with a lower access level calls this method, the server SHOULD return ERROR_ACCESS_DENIED.
- The WINS server retrieves the entries from the specified file and registers the retrieved names into the WINS database.
- After the WINS server finishes the initialization, it removes the file if *fDel* is set to a nonzero value.
- The WINS server SHOULD return ERROR_STATIC_INIT_FAILED if any error occurs while the server is reading the file or registering the names in the database.

3.1.4.5 R_WinsDoScavenging (Opnum 4)

The R_WinsDoScavenging method queues a **scavenging** request on the **target WINS server**.

```
DWORD R_WinsDoScavenging(  
    [in] handle_t ServerHdl  
);
```

ServerHdl: An **RPC** binding over IP address/HostName to the **WINS** server. RPC uses this binding internally to determine which WINS server the call is directed to.

Return Values: A 32 bit unsigned integer that indicates the return status. A return value of ERROR_SUCCESS (0x00000000) indicates that the operation completed successfully. A nonzero return value is a Win32 error code, as specified in [\[MS-ERREF\]](#). The following Win32 error codes can be returned:

Return value/code	Description
0x00000000 ERROR_SUCCESS	The call was successful.
0x00000005 ERROR_ACCESS_DENIED	The caller doesn't have sufficient permissions.

Exceptions Thrown: No exceptions are thrown beyond those thrown by the underlying RPC protocol [\[MS-RPCE\]](#).

Processing and Response Requirements:

When R_WinsDoScavenging is called, the server returns immediately without waiting for scavenging to start. The server just queues a request for the scavenging operation, and the internal state and configuration of the WINS server determine whether or not the scavenging occurs. Hence, callers to R_WinsDoScavenging SHOULD NOT treat a return code of ERROR_SUCCESS as indicating a successful scavenging operation. Instead, callers SHOULD rely on WINS event logs to determine whether or not the scavenging operation succeeded. The following table lists the events that indicate the status of scavenging.

Event ID	Event Name	Event Description
Informational events		
4143	WINS_EVT_SCV_RECS	WINS scavenged its records in the WINS database. The number of records scavenged is given in the data section.
4144	WINS_EVT_SCV_RPLTOMB	WINS scavenged replica tombstones in the WINS database. The number of records scavenged is given in the data section.
4247	WINS_EVT_SCV_EXC	The WINS Scavenger thread encountered an error.
4250	WINS_EVT_SCV_ERR	The WINS Scavenger thread could not scavenge a record. This record will be ignored. The Scavenger will continue to the next available record. Check the application log for the Exchange component, ESENT.
4269	WINS_EVT_UNABLE_TO_CHG_PRIORITY	The Scavenger thread was unable to change its priority level.
4288	WINS_EVT_CLEANUP_OWNADDTBL_EXC	The Scavenger thread encountered an error while cleaning up the owner-address table. It will try again after the Verify interval has elapsed.
4328	WINS_EVT_ADMIN_SCVENGING_INITIATED	Administrator '%1' has initiated a scavenging operation.
4329	WINS_EVT_SCVENGING_STARTED	The WINS server has started a scavenging operation.
4330	WINS_EVT_SCVENGING_COMPLETED	The WINS server has completed the scavenging operation.

Event ID	Event Name	Event Description
5001	WINS_EVT_SCV_RANGE	WINS is scavenging the locally owned records from the database. The version number range that is scavenged is given in the data section, in the second to fifth words, in the order: from_version_number (low word, high word) to_version_number (low word, high word).
5002	WINS_EVT_SCV_CHUNK	WINS is scavenging a chunk of N records in the version number range from X to Y. N, X and Y (low word, high word for version numbers) are given in the second to sixth words in the data section.
Warning events		
4150	WINS_EVT_ADJ_TIME_INTVL_R	WINS adjusted the scavenging-related time interval, %1, so that it is compatible with the replication time intervals. The adjusted value for this scavenging parameter is given in the data section (second DWORD). This value was computed by WINS using an algorithm that MAY use the maximum replication time interval specified. The current value achieves a good balance between consistency of databases across the network of WINS servers and the performance of the WINS servers.
4151	WINS_EVT_ADJ_TIME_INTVL	WINS adjusted the scavenging-related time interval, %1. The adjusted value for this scavenging parameter is given in the data section (second DWORD). This value was computed by WINS using an algorithm that tries to achieve a good balance between consistency of databases across the network of WINS servers and the performance of the WINS servers.
4153	WINS_EVT_UNABLE_TO_VERIFY	The Scavenger thread found active replicas that needed to be verified with the owning WINS server because they were older than the verify time interval. The table of owner-to-address mappings indicated the WINS was not active.
4164	WINS_EVT_FORCE_SCV_R_T	WINS was forced to scavenge replica tombstones of a WINS. The person with administrative rights on the computer forced the scavenging by using the winscl.exe. WINS does not scavenge replica tombstones unless they have timed out and WINS has been running for at least three days (this ensures that the tombstones have replicated to other WINS). In this case, the tombstones were timed out but the WINS had not been running for three days. The replica tombstones were deleted. This deletion does not constitute a problem unless there are WINS servers that are primary and backup to clients but not both Push and Pull partners of each other. With the preceding WINS scenario, there is a low probability that this action will result in database inconsistency but if it does, a consistent state can be achieved by initiating consistency checks by using the winscl.exe.

The following requirements and recommendations apply to a WINS server that processes a call to [R_WinsDoScavenging](#):

- Callers to R_WinsDoScavenging SHOULD have control level access. If an RPC client with a lower access level calls R_WinsDoScavenging, the server SHOULD return ERROR_ACCESS_DENIED.

- The WINS server queues a request on the target WINS server for the scavenging operation, and the method returns immediately with `ERROR_SUCCESS` as the status code.

3.1.4.6 R_WinsGetDbRecs (Opnum 5)

The `R_WinsGetDbRecs` method returns the records whose version numbers are within a specified range and that are owned by a specified **WINS server**.

```
DWORD R_WinsGetDbRecs(
    [in] handle_t ServerHdl,
    [in, ref] PWINSINTF_ADD_T pWinsAdd,
    [in] WINSINTF_VERS_NO_T MinVersNo,
    [in] WINSINTF_VERS_NO_T MaxVersNo,
    [out] PWINSINTF_RECS_T pRecs
);
```

ServerHdl: An **RPC** binding over IP address/HostName to the WINS server. RPC uses this binding to resolve which WINS server the call is directed to.

pWinsAdd: Address of an **owner WINS server** whose records are retrieved from the **target WINS server**.

MinVersNo: The lower bound on the version range of the records to be retrieved.

MaxVersNo: The upper bound on the version range of the records to be retrieved.

pRecs: Pointer to a structure of type [WINSINTF_RECS_T](#), which contains the records retrieved from the target WINS server.

Return Values: A 32-bit unsigned integer value that indicates the return status. A return value of `ERROR_SUCCESS` (0x00000000) indicates that the operation finished successfully. Any nonzero value is a Win32 error code, as specified in [\[MS-ERREF\]](#). The following Win32 error codes can be returned:

Return value/code	Description
0x00000000 <code>ERROR_SUCCESS</code>	The call was successful.
0x00000FA0 <code>ERROR_WINS_INTERNAL</code>	An error occurred while processing the RPC call.
0x00000005 <code>ERROR_ACCESS_DENIED</code>	The caller does not have sufficient permissions.

Exceptions Thrown: No exceptions SHOULD be thrown beyond those thrown by the underlying RPC protocol [\[MS-RPCE\]](#).

Processing and Response Requirements:

The following requirements and recommendations apply to a WINS server that processes a call to [R_WinsGetDbRecs](#):

- The RPC method caller SHOULD have query-level access. <7> If an RPC client with a lower access level calls `R_WinsGetDbRecs`, the server SHOULD return `ERROR_ACCESS_DENIED`.
- In response to a `R_WinsGetDbRecs` call, records are retrieved from the target WINS server database if their version numbers fall between *MinVersNo* and *MaxVersNo*, and if the records are owned by the owner WINS server whose address is specified by *pWinsAdd*.

- If the `R_WinsGetDbRecs` caller specifies zero for both *MinVersNo* and *MaxVersNo*, all records owned by the WINS server specified by *pWinsAdd* are retrieved from the target WINS server's database.
- The *MinVersNo* value MUST be less than or equal to *MaxVersNo* value for the `R_WinsGetDbRecs` call to succeed; otherwise, the server SHOULD return `ERROR_WINS_INTERNAL`.

The `R_WinsGetDbRecs` caller is responsible for freeing the memory pointed to by *pRecs->pRow->pName* and *pRecs->pRow->pAdd* for each record, then using the **midl_user_free** function (section 3) to free the *pRecs->pRow* and *pRecs* pointers themselves.

3.1.4.7 R_WinsTerm (Opnum 6)

The `R_WinsTerm` method sends a termination signal to the **WINS** process on a **target WINS server**.

```
DWORD R_WinsTerm(
    [in] handle_t ServerHdl,
    [in] SHORT fAbruptTem
);
```

ServerHdl: An **RPC** binding over IP address/HostName to the WINS server. RPC uses this binding internally to determine which WINS server the call is directed to.

fAbruptTem: A value that indicates whether the WINS process terminates immediately. If this value is nonzero, the service terminates immediately. Otherwise, the service exits normally and frees all resources.

Return Values: A 32 bit unsigned integer that indicates the return status. A return value of `ERROR_SUCCESS` (0x00000000) indicates that operation completed successfully. A nonzero return value is a Win32 error code, as specified in [\[MS-ERREF\]](#). The following Win32 error codes can be returned:

Return value/code	Description
0x00000000 ERROR_SUCCESS	The call was successful.
0x00000005 ERROR_ACCESS_DENIED	The caller doesn't have sufficient permissions.

Exceptions Thrown: No exceptions SHOULD be thrown beyond those thrown by the underlying RPC protocol [\[MS-RPCE\]](#).

Processing and Response Requirements:

The following requirements and recommendations apply to a WINS server that processes a call to [R_WinsTerm](#):

- The `R_WinsTerm` caller SHOULD have control level access. If an RPC client with a lower access level calls `R_WinsTerm`, the server SHOULD return `ERROR_ACCESS_DENIED`.
- `R_WinsTerm` always returns `ERROR_SUCCESS` if the client has sufficient access level permissions.
- If *fAbruptTem* is set to a nonzero value, the service exits immediately. Otherwise, the service frees all the resources and then calls the exit process.

3.1.4.8 R_WinsBackup (Opnum 7)

The R_WinsBackup method backs up the **WINS** database to a specified directory.

```
DWORD R_WinsBackup(  
    [in] handle_t ServerHdl,  
    [in, string, ref] LPBYTE pBackupPath,  
    [in] SHORT fIncremental  
);
```

ServerHdl: An **RPC** binding over IP address/HostName to the WINS server. RPC uses this binding internally to determine which WINS server the call is directed to.

pBackupPath: A pointer to a string that contains the name of the directory to which to back up the database. This pointer MUST not be NULL.

fIncremental: A value that is ignored.

Return Values: A 32-bit unsigned integer that indicates the return status. A return value of ERROR_SUCCESS (0x00000000) indicates that the operation completed successfully. Any nonzero return value is a Win32 error code, as specified in [\[MS-ERREF\]](#). The following Win32 error codes can be returned.

Return value/code	Description
0x00000000 ERROR_SUCCESS	The call was successful.
0x00000FA0 ERROR_WINS_INTERNAL	An error occurred while processing the RPC call.
0x00000FA4 ERROR_FULL_BACKUP	The backup failed. Check the directory to which you are backing up the database.
0x00000005 ERROR_ACCESS_DENIED	The caller doesn't have sufficient permissions.

Exceptions Thrown: No exceptions are thrown beyond those thrown by the underlying RPC protocol [\[MS-RPCE\]](#).

Processing and Response Requirements:

The following requirements and recommendations apply to a WINS server that processes a call to [R_WinsBackup](#):

- The R_WinsBackup caller SHOULD have control level access. If an RPC client with a lower access level calls this method, the server SHOULD return ERROR_ACCESS_DENIED.
- The server returns ERROR_WINS_INTERNAL if *pBackupPath* points to a string that is longer than 255 characters.
- The database is always backed up to the path specified by *pBackupPath* with the string "\wins_bak\" appended. If the client doesn't have sufficient permissions to create files in the specified directory or if the backup fails for any other reasons, the server SHOULD return an ERROR_FULL_BACKUP error.

3.1.4.9 R_WinsDelDbRecs (Opnum 8)

The `R_WinsDelDbRecs` method deletes the records whose version numbers are within a specified range and that are owned by a specified **WINS server**.

```
DWORD R_WinsDelDbRecs(  
    [in] handle_t ServerHdl,  
    [in, ref] PWINSINTF_ADD T pWinsAdd,  
    [in] WINSINTF_VERS_NO_T MinVersNo,  
    [in] WINSINTF_VERS_NO_T MaxVersNo  
);
```

ServerHdl: An **RPC** binding over IP address/HostName to the WINS server. RPC uses this binding internally to determine which WINS server the call is directed to.

pWinsAdd: A pointer to an **owner WINS server** address whose records are to be deleted from the **target WINS server**.

MinVersNo: The lower bound on the version number of the records to be deleted.

MaxVersNo: The upper bound on the version number of the records to be deleted.

Return Values: A 32-bit unsigned integer that indicates the return status. A return value of `ERROR_SUCCESS` (0x00000000) indicates that the operation completed successfully. Any nonzero return value is a Win32 error code, as specified in [\[MS-ERREF\]](#). The following Win32 error codes can be returned.

Return value/code	Description
0x00000000 ERROR_SUCCESS	The call was successful.
0x00000FA0 ERROR_WINS_INTERNAL	An error occurred while processing the RPC call.
0x00000005 ERROR_ACCESS_DENIED	The caller doesn't have sufficient permissions.

Exceptions Thrown: No exceptions are thrown beyond those thrown by the underlying RPC protocol [\[MS-RPCE\]](#).

Processing and Response Requirements:

The following requirements and recommendations apply to a WINS server that processes a call to [R_WinsDelDbRecs](#):

- The `R_WinsDelDbRecs` caller SHOULD have control-level access. If an RPC client with a lower access level calls this method, the server SHOULD return `ERROR_ACCESS_DENIED`.
- If the target WINS server doesn't have any records owned by the WINS server whose address is specified by `pWinsAdd`, the server SHOULD return `ERROR_WINS_INTERNAL`.
- Records are deleted from the target WINS server database if their version numbers fall between the values of `MinVersNo` and `MaxVersNo` and if they are owned by the WINS server whose address is specified in `pWinsAdd`.
- If both `MinVersNo` and `MaxVersNo` are set to zero, all records owned by the WINS server whose address is specified in `pWinsAdd` are deleted.

3.1.4.10 R_WinsPullRange (Opnum 9)

The R_WinsPullRange method pulls a range of records owned by a **WINS** server from another WINS server, and replicates them within the **target WINS server** database. <8>

```
DWORD R_WinsPullRange(  
    [in] handle_t ServerHdl,  
    [in, ref] PWINSINTF_ADD T pWinsAdd,  
    [in, ref] PWINSINTF_ADD_T pOwnerAdd,  
    [in] WINSINTF_VERS_NO_T MinVersNo,  
    [in] WINSINTF_VERS_NO_T MaxVersNo  
);
```

ServerHdl: An **RPC** binding over IP address/HostName to the WINS server. RPC uses this binding internally to determine which WINS server the call is directed to.

pWinsAdd: The address of the WINS server from which the entries are pulled.

pOwnerAdd: The address of the **owner WINS server** whose entries are pulled.

MinVersNo: The lower bound on the range of version numbers for the records to be pulled.

MaxVersNo: The upper bound on the range of version numbers for the records to be pulled.

Return Values: A 32 bit unsigned integer that indicates the return status. A return value of ERROR_SUCCESS (0x00000000) indicates that operation completed successfully. A nonzero return value is a Win32 error code, as specified in [\[MS-ERREF\]](#). The following Win32 error codes can be returned:

Return value/code	Description
0x00000000 ERROR_SUCCESS	The call was successful.
0x00000FA0 ERROR_WINS_INTERNAL	An error occurred while processing the RPC call.
0x00000005 ERROR_ACCESS_DENIED	The caller doesn't have sufficient permissions.

Exceptions Thrown: No exceptions are thrown beyond those thrown by the underlying RPC protocol [\[MS-RPCE\]](#).

Processing and Response Requirements:

When R_WinsPullRange is called, the server returns immediately without waiting for the actual pull. It just queues a request for the pull operation, and the actual pull starts at a time determined by the current state and configuration of the target WINS server. Hence, R_WinsPullRange callers SHOULD NOT treat an ERROR_SUCCESS return value as indicating a successful pull operation. Instead, callers SHOULD rely on WINS event logs to determine whether or not the pull operation succeeded. The following table lists the events that indicate the status of a pull operation.

Event ID	Event Name	Event Description
Informational events		
4104	WINS_EVT_NO_PULL_RECS	There are no pull records.

Event ID	Event Name	Event Description
4124	WINS_EVT_UPD_NTF_NOT_ACCEPTED	The WINS server received an update notification from the nonconfigured WINS server at the address, %1. The WINS server rejected it. This means the remote WINS server is not in the list of Push partners (WINS servers under the Pull key) and the administrator has prohibited (by using the registry) replication with nonconfigured WINS servers. To have this WINS server accept update notifications from nonconfigured WINS servers, set the Wins\Parameters\RplOnlyWCnfPnrs value in the registry to zero.
4126	WINS_EVT_ADD_VERS_MAP_REQ_NOT_ACCEPTED	The WINS server received a pull request from the nonconfigured WINS server with the address, %1. The WINS server rejected it since the remote WINS server is not in the list of Pull partners (WINS servers under the Pull key) and the administrator has prohibited (using the registry) replication with nonconfigured partners. If you want this WINS server to accept update notifications from WINS servers not in the pull partner list, set the "replication only with configured partners" value in the registry to zero.
4141	WINS_EVT_REC_PULLED	WINS pulled records from a WINS while doing %1. The partner's address and the owner's address whose records were pulled are in the data section (second and third DWORD respectively). The number of records pulled is the fourth DWORD.
4142	WINS_EVT_CC_NO_RECS	WINS performed a consistency check on the records. The number of records pulled, the address of the WINS whose records were pulled, and the address of the WINS from which these records were pulled are given in the second, third, and fourth DWORDs in the data section.
4231	WINS_EVT_CANT_QUERY_PULL_KEY	WINS could not get information about the Pull key. Check whether the permissions on the key are set properly, system resources are low, or the registry is having a problem.
4235	WINS_EVT_CANT_OPEN_PULL_SUBKEY	WINS could not open a Pull subkey. Check whether the permissions on the key are set properly, system resources are low, or the registry is having a problem.
4237	WINS_EVT_CANT_GET_PULL_TIMEINT	WINS could not get the time interval from a Pull record.
4243	WINS_EVT_RPLPULL_PUSH_NTF_EXC	WINS Pull thread encountered an error during the process of sending a push notification to another WINS. The error code is in the data section.
4255	WINS_EVT_PUSH_PNR_INVALID_ADD	WINS has been asked to pull its entries. Check all the Pull subkeys of this WINS.
4273	WINS_EVT_PULL_RANGE_EXC	An error was encountered while trying to service a pull range request from a remote WINS. The exception code is the second DWORD of the data section.

Event ID	Event Name	Event Description
4284	WINS_EVT_EXC_PULL_TRIG_PROC	WINS encountered an exception while processing a pull trigger.
Warning events		
4161	WINS_EVT_PARTIAL_RPL_TYPE	A nonzero replication type applies for this partner. This means only a subset of records will be replicated between the local WINS and this partner. To get records that did not replicate, either pull them by using the winscl.exe in the Windows 2000 Resource Kit (as described in [MSFT-ResourceKits]), or delete all owners acquired only through this partner and then initiate replication to reacquire all their records. The partner's address is given in the second DWORD of the data section.
Error events		
4178	WINS_EVT_CANT_OPEN_PULL_KEY	The WINS Pull configuration key could not be created or opened. Check to see if the permissions on the key are set properly, system resources are low, or the registry is having a problem.
4194	WINS_EVT_CANT_CREATE_NTF SOCK	WINS could not create the User Datagram Protocol (UDP) socket to listen for Connection notification messages sent by another Pull thread in the local WINS.

The following requirements and recommendations apply to a WINS server that processes a call to R_WinsPullRange:

- R_WinsPullRange callers SHOULD have control level access. If an RPC client with a lower access level calls this method, the server SHOULD return ERROR_ACCESS_DENIED.
- The value of *MinVersNo* MUST be less than or equal to the value of *MaxVersNo*. Otherwise, the server SHOULD return ERROR_WINS_INTERNAL.
- If the target WINS server is configured to pull records only from configured partners, the WINS server address given in *pWinsAdd* MUST have been configured as a pull partner for the target WINS server. Otherwise, the server SHOULD return ERROR_WINS_INTERNAL.
- When the client queues a request to pull the records owned by the server whose address is given in *pOwnerAdd* from the WINS server whose address is given in *pWinsAdd*, the RPC call SHOULD return immediately without waiting for the replication operation to complete.

3.1.4.11 R_WinsSetPriorityClass (Opnum 10)

The R_WinsSetPriorityClass method sets the **priority class** for the **WINS** process running on the **target WINS server**.

```
DWORD R_WinsSetPriorityClass(
    [in] handle t ServerHdl,
    [in] WINSINTF_PRIORITY_CLASS_E PrCls_e
);
```


ServerHdl: An **RPC** binding over IP address/HostName to the WINS server. RPC uses this binding internally to determine which WINS server the call is directed to.

PrCls_e: The priority class to be set.

Return Values: A 32 bit unsigned integer that indicates the return status. A return value of ERROR_SUCCESS (0x00000000) indicates that the operation completed successfully. A nonzero return value is a Win32 error code, as specified in [\[MS-ERREF\]](#). The following Win32 error codes can be returned:

Return value/code	Description
0x00000000 ERROR_SUCCESS	The call was successful.
0x00000FA0 ERROR_WINS_INTERNAL	An error occurred while processing the RPC call.
0x00000005 ERROR_ACCESS_DENIED	The caller doesn't have sufficient permissions.

Exceptions Thrown: No exceptions are thrown beyond those thrown by the underlying RPC protocol [\[MS-RPCE\]](#).

Processing and Response Requirements:

The following requirements and recommendations apply to a WINS server that processes a call to [R_WinsSetPriorityClass](#):

- The R_WinsSetPriorityClass caller SHOULD have control level access. If an RPC client with a lower access level calls this method, the server SHOULD return ERROR_ACCESS_DENIED.
- If PrCls_e is set to a value other than WINSINTF_E_NORMAL or WINSINTF_E_HIGH, the server SHOULD return ERROR_WINS_INTERNAL.
- The server sets the priority class of the WINS process to the one specified by PrsCls_e.

3.1.4.12 R_WinsResetCounters (Opnum 11)

The R_WinsResetCounters method resets the pull **replication** counters for all partners of the **target WINS server**.

```
DWORD R_WinsResetCounters(  
    [in] handle t ServerHdl  
);
```

ServerHdl: An **RPC** binding over IP address/HostName to the **WINS** server. RPC uses this binding internally to determine which WINS server the call is directed to.

Return Values: A 32 bit unsigned integer that indicates the return status. A return value of ERROR_SUCCESS (0x00000000) indicates that operation completed successfully. A nonzero return value is a Win32 error code, as specified in [\[MS-ERREF\]](#). The following Win32 error codes can be returned:

Return value/code	Description
0x00000000	The call was successful.

Return value/code	Description
ERROR_SUCCESS	
0x00000005 ERROR_ACCESS_DENIED	The caller doesn't have sufficient permissions.

Exceptions Thrown: No exceptions are thrown beyond those thrown by the underlying RPC protocol [\[MS-RPCE\]](#).

Processing and Response Requirements:

The following requirements and recommendations apply to a WINS server that processes a call to [R_WinsResetCounters](#):

- The R_WinsResetCounters caller SHOULD have control level access. If an RPC client with a lower access level calls this method, the server SHOULD return ERROR_ACCESS_DENIED.
- Each WINS server maintains one [WINSINTF_RPL_COUNTERS_T](#) structure (section 2.2.2.5) per configured **pull partner** to track the number of successful pull replications and the number of communication failures. The R_WinsResetCounters method resets to zero the values of the **NoOfRpls** and **NoOfCommFails** members of the WINSINTF_RPL_COUNTERS_T structures for all the configured pull partners of the target WINS server.
- This method MUST return ERROR_SUCCESS if the client has sufficient access level permissions.

3.1.4.13 R_WinsWorkerThdUpd (Opnum 12)

The R_WinsWorkerThdUpd method updates the number of threads that have been created to serve **NetBIOS** requests.

```
DWORD R_WinsWorkerThdUpd(
    [in] handle_t ServerHdl,
    [in] DWORD NewNoOfNbtThds
);
```

ServerHdl: An **RPC** binding over IP address/HostName to the **WINS** server. RPC uses this binding internally to determine which WINS server the call is directed to.

NewNoOfNbtThds: New value for the number of worker threads that have been created for NetBIOS requests.

Return Values: A 32 bit unsigned integer that indicates the return status. A return value of ERROR_SUCCESS (0x00000000) indicates that operation completed successfully. A nonzero return value is a Win32 error code, as specified in [\[MS-ERREF\]](#). The following Win32 error codes can be returned:

Return value/code	Description
0x00000000 ERROR_SUCCESS	The call was successful.
0x00000FA0 ERROR_WINS_INTERNAL	An error occurred while processing the RPC call.
0x00000005 ERROR_ACCESS_DENIED	The caller doesn't have sufficient permissions.

Exceptions Thrown: No exceptions are thrown beyond those thrown by the underlying RPC protocol [\[MS-RPCE\]](#).

Processing and Response Requirements:

The following requirements and recommendations apply to a WINS server that processes a call to [R_WinsWorkerThdUpd](#):

- The R_WinsWorkerThdUpd caller SHOULD have control level access. If an RPC client with a lower access level calls this method, the server SHOULD return ERROR_ACCESS_DENIED.
- The WINS service MUST be in the running or paused state for this method to succeed. If the service is in the initializing or exiting state, the server SHOULD return ERROR_WINS_INTERNAL.
- The new number given in *NewNoOfNbtThds* MUST be in the range 2 through 19, inclusive. Otherwise, the server SHOULD return an ERROR_WINS_INTERNAL error.
- The R_WinsWorkerThdUpd call sets the number of worker threads that serve NetBIOS requests to the new number given in *NewNoOfNbtThds*. If the existing number of NetBIOS threads is same as the requested number, the RPC call SHOULD return immediately. Otherwise, NetBIOS threads are created or deleted to adjust the total number of threads to the requested number.

3.1.4.14 R_WinsGetNameAndAdd (Opnum 13)

The R_WinsGetNameAndAdd method retrieves the **NetBIOS** name and the corresponding IP address of the **target WINS server**.

```
DWORD R_WinsGetNameAndAdd(  
    [in] _handle_t ServerHdl,  
    [out, ref] PWINSINTF_ADD_T pWinsAdd,  
    [out, string, size_is(80)] LPBYTE pUncName  
);
```

ServerHdl: An **RPC** binding over IP address/HostName to the **WINS** server. RPC uses this binding internally to determine which WINS server the call is directed to.

pWinsAdd: A pointer to a structure containing the IP address of the target WINS server.

pUncName: A pointer to a NULL-terminated string containing the NetBIOS name of the target WINS server.

Return Values: A 32 bit unsigned integer that indicates the return status. A return value of ERROR_SUCCESS (0x00000000) indicates that operation completed successfully. A nonzero return value is a Win32 error code, as specified in [\[MS-ERREF\]](#). The following Win32 error codes can be returned:

Return value/code	Description
0x00000000 ERROR_SUCCESS	The call was successful.
0x00000005 ERROR_ACCESS_DENIED	The caller doesn't have sufficient permissions.

Exceptions Thrown: No exceptions are thrown beyond those thrown by the underlying RPC protocol [\[MS-RPCE\]](#).

Processing and Response Requirements:

The following requirements and recommendations apply to a WINS server that processes a call to [R_WinsGetNameAndAdd](#):

- The R_WinsGetNameAndAdd caller SHOULD have query-level access. [<9>](#) If an RPC client with a lower access level calls this method, the server SHOULD return ERROR_ACCESS_DENIED. [<10>](#)
- The structure that *pWinsAdd* points to contains only an IP address. The R_WinsGetNameAndAdd caller SHOULD ignore the other fields of the structure.
- The server retrieves the NetBIOS name by calling a standard Windows function, which returns the status code directly to the caller without any modification. Hence, any Win32 error code can be returned, as specified in [MS-ERREF].

3.1.4.15 R_WinsGetBrowserNames_Old (Opnum 14)

The R_WinsGetBrowserNames_Old method always returns an ERROR_WINS_INTERNAL error code.

```
DWORD R_WinsGetBrowserNames_Old(  
    [in] handle_t ServerHdl,  
    [out] PWINSINTF_BROWSER_NAMES_T pNames  
);
```

ServerHdl: An **RPC** binding over IP address/HostName to the **WINS** server. RPC uses this binding internally to determine which WINS server the call is directed to.

pNames: This field MUST be ignored.

Return Values: A 32-bit unsigned integer value that indicates the return status. The method always returns the ERROR_WINS_INTERNAL error code.

Return value/code	Description
0x00000FA0 ERROR_WINS_INTERNAL	An error occurred while processing the RPC call.

Exceptions Thrown: No exceptions are thrown.

Processing and Response Requirements:

Clients with any access level can call this method.

3.1.4.16 R_WinsDeleteWins (Opnum 15)

The R_WinsDeleteWins method deletes all the records owned by a particular **WINS server** from the **target WINS server** database.

```
DWORD R_WinsDeleteWins(  
    [in] handle_t ServerHdl,  
    [in, ref] PWINSINTF_ADD_T pWinsAdd  
);
```

ServerHdl: An **RPC** binding over IP address/HostName to the WINS server. RPC uses this binding internally to determine which WINS server the call is directed to.

pWinsAdd: A pointer to the address of the **owner WINS server** whose records are to be deleted from the target WINS server.

Return Values: A 32 bit unsigned integer that indicates the return status. A return value of ERROR_SUCCESS (0x00000000) indicates that operation completed successfully. A nonzero return value is a Win32 error code, as specified in [\[MS-ERREF\]](#). The following Win32 error codes can be returned:

Return value/code	Description
0x00000000 ERROR_SUCCESS	The call was successful.
0x00000FA0 ERROR_WINS_INTERNAL	An error occurred while processing the RPC call.
0x00000005 ERROR_ACCESS_DENIED	The caller doesn't have sufficient permissions.

Exceptions Thrown: No exceptions are thrown beyond those thrown by the underlying RPC protocol [\[MS-RPCE\]](#).

Processing and Response Requirements:

The following requirements and recommendations apply to a WINS server that processes a call to [R_WinsDeleteWins](#):

- The RPC method caller SHOULD have control-level access. If an RPC client with a lower access level calls this method, the server SHOULD return ERROR_ACCESS_DENIED.
- If *pWinsAdd* contains the IP address of the target **WINS**, the records are deleted immediately from the target WINS server database. If the server encounters an error while retrieving the records from the database, it SHOULD return ERROR_WINS_INTERNAL; otherwise, the server returns ERROR_SUCCESS.
- If *pWinsAdd* contains an IP address different from the target WINS server address, a request is queued at the target WINS server, and the RPC call returns immediately with ERROR_SUCCESS status.

3.1.4.17 R_WinsSetFlags (Opnum 16)

The R_WinsSetFlags method always returns ERROR_SUCCESS.

```
DWORD R_WinsSetFlags(  
    [in] handle t ServerHdl,  
    [in] DWORD fFlags  
);
```

ServerHdl: An **RPC** binding over IP address/HostName to the **WINS** server. RPC uses this binding internally to determine which WINS server the call is directed to.

fFlags: This field MUST be ignored.

Return Values: A 32-bit unsigned integer value that indicates the return status. A return value of ERROR_SUCCESS (0x00000000) indicates that the operation completed successfully.

Return value/code	Description
0x00000000 ERROR_SUCCESS	The call was successful.

Exceptions Thrown: No exceptions are thrown.

Processing and Response Requirements:

Clients with any access level can call this method.

3.1.4.18 R_WinsGetBrowserNames (Opnum 17)

The R_WinsGetBrowserNames method retrieves **browser name** records from the **target WINS server** database.

```
DWORD R_WinsGetBrowserNames(  
    [in, ref] WINSIF_HANDLE ServerHdl,  
    [out] PWINSINTF_BROWSER_NAMES_T pNames  
);
```

ServerHdl: An **RPC** binding over IP address/HostName to the **WINS server**. RPC uses this binding internally to determine which WINS server the call is directed to. See [\[MSDN-Handles\]](#) for more information.

This value MUST be ignored by the WINS server on receipt.

pNames: A pointer to a structure of type [WINSINTF_BROWSER_NAMES_T \(section 2.2.2.10\)](#), which contains the browser name records retrieved from the target WINS server.

Return Values: A 32-bit unsigned integer that indicates the return status. A return value of ERROR_SUCCESS (0x00000000) indicates that operation completed successfully. A nonzero return value is a Win32 error code, as specified in [\[MS-ERREF\]](#). The following Win32 error codes can be returned:

Return value/code	Description
0x00000000 ERROR_SUCCESS	The call was successful.
0x00000FA0 ERROR_WINS_INTERNAL	An error occurred while processing the RPC call.

Exceptions Thrown: No exceptions SHOULD be thrown beyond those thrown by the underlying RPC protocol [\[MS-RPCE\]](#).

Processing and Response Requirements:

Clients with any access level can call this method.

The following requirements and recommendations apply to a WINS server that processes a call to [R_WinsGetBrowserNames](#):

- This method retrieves all browser name records in the target WINS server database.
- If the **Browser name cache** abstract data element (section [3.1.1](#)) has been populated, and less than 3 minutes have elapsed since it was last updated, this method SHOULD return the records from the cache by using the *pNames* parameter.
- If this method call is being made for the first time, or if 3 minutes or more have elapsed since the **Browser name cache** was last updated, the cache SHOULD be refreshed by fetching records from the database, and the contents of the cache are returned.

- If any error occurs while retrieving the records, the service SHOULD return an ERROR_WINS_INTERNAL error code.

The R_WinsGetBrowserNames caller is responsible for freeing the memory pointed to by *pRecs->pRow->pName* and *pRecs->pRow->pAdd* for each record, then using the **midl_user_free** function (section 3) to free the *pRecs->pRow* and *pRecs* pointers themselves.

3.1.4.19 R_WinsGetDbRecsByName (Opnum 18)

The R_WinsGetDbRecsByName method retrieves records matching an **owner** address from a **target WINS server** database starting at a specified cursor.

```
DWORD R_WinsGetDbRecsByName (
    [in] handle t ServerHdl,
    [in, unique] PWINSINTF_ADD_T pWinsAdd,
    [in] DWORD Location,
    [in, unique, size is (NameLen + 1)]
    LPBYTE pName,
    [in] DWORD NameLen,
    [in] DWORD NoOfRecsDesired,
    [in] DWORD fOnlyStatic,
    [out] PWINSINTF_RECS_T pRecs
);
```

ServerHdl: An **RPC** binding over IP address/HostName to the **WINS server**. RPC uses this binding internally to determine which WINS server the call is directed to.

pWinsAdd: A pointer to the address of the **owner WINS server** whose records are to be retrieved. If the pointer is NULL, the records for all owners are retrieved.

Location: A value specifying the direction in which the database is searched. If the value is zero, the database is searched forward starting from the beginning. If the value is 1, the database is searched backward starting from the last record of the database.

pName: A pointer to a name that specifies the cursor from which the database retrieval starts.

NameLen: The length of the name that *pName* points to, including terminating NULL character.

NoOfRecsDesired: The number of records to be retrieved from the database.

fOnlyStatic: Takes a value of 1, 2, or 4 to indicate whether **static records**, **dynamic records**, or both are retrieved. A value of 1 retrieves only static records. A value of 2 retrieves only dynamic records. A value of 4 retrieves both static records and dynamic records.

pRecs: A pointer to a structure containing the retrieved records.

Return Values: A 32 bit unsigned integer that indicates the return status. A return value of ERROR_SUCCESS (0x00000000) indicates that operation completed successfully. A nonzero return value is a Win32 error code, as specified in [\[MS-ERREF\]](#). The following Win32 error codes can be returned:

Return value/code	Description
0x00000000 ERROR_SUCCESS	The call was successful.
0x00000FA0 ERROR_WINS_INTERNAL	An error occurred while processing the RPC call.

Return value/code	Description
0x00000FA5 ERROR_REC_NON_EXISTENT	No records were found matching the given data.
0x00000005 ERROR_ACCESS_DENIED	The caller doesn't have sufficient permissions.

Exceptions Thrown: No exceptions SHOULD be thrown beyond those thrown by the underlying RPC protocol [\[MS-RPCE\]](#).

Processing and Response Requirements:

The following requirements and recommendations apply to a WINS server that processes a call to [R_WinsGetDbRecsByName](#):

- The RPC method caller SHOULD have query-level access. [<11>](#) If an RPC client with a lower access level calls this method, the server SHOULD return ERROR_ACCESS_DENIED.
- This method returns all records whose owner address matches the address specified in *pWinsAdd*. If *pName* points to a valid name, the database search starts from the record after the record whose name matches the valid name. If the name that *pName* points to does not match the name for any database records, the database search starts from the beginning of the database.
- A maximum of 5,000 records can be retrieved in a single call.
- If the owner's address is specified and if the server can't find this address in its **owner version map**, the server returns error ERROR_WINS_INTERNAL error.
- If no records match the search criteria, the server returns an ERROR_REC_NON_EXISTENT error. For any other error conditions, the server returns an ERROR_WINS_INTERNAL error.
- Refer to [Retrieving All the Records of a WINS Database \(section 4.6\)](#) to see how to use [R_WinsGetDbRecsByName](#) to retrieve all the records of a database.

The [R_WinsGetDbRecsByName](#) caller is responsible for freeing the memory pointed to by *pRecs->pRow->pName* and *pRecs->pRow->pAdd* for each record, then using the **midl_user_free** function (section [3](#)) to free the *pRecs->pRow* and *pRecs* pointers themselves.

3.1.4.20 R_WinsStatusNew (Opnum 19)

The [R_WinsStatusNew](#) method retrieves configuration settings and statistics from a **WINS** server.

```
DWORD R_WinsStatusNew(
    [in] handle_t ServerHdl,
    [in] WINSINTF_CMD_E Cmd_e,
    [out] PWINSINTF_RESULTS_NEW T pResults
);
```

ServerHdl: An **RPC** binding over IP address/HostName to the WINS server. RPC uses this binding internally to determine which WINS server the call is directed to.

Cmd_e: The command to be executed on the **target WINS server**, from the [WINSINTF_CMD_E](#) enumeration (section 2.2.1.5).

pResults: A pointer to a [WINSINTF_RESULTS_NEW T](#) structure (section 2.2.2.11), which contains the results of the command execution.

Return Values: A 32-bit unsigned integer that indicates the return status. A return value of ERROR_SUCCESS (0x00000000) indicates that operation completed successfully. A nonzero return value is a Win32 error code, as specified in [\[MS-ERREF\]](#). The following Win32 error codes can be returned:

Return value/code	Description
0x00000000 ERROR_SUCCESS	The call was successful.
0x00000005 ERROR_ACCESS_DENIED	The caller does not have sufficient permissions.
0x00000FA0 ERROR_WINS_INTERNAL	An error occurred while processing the RPC call.

Exceptions Thrown: No exceptions SHOULD be thrown beyond those thrown by the underlying RPC protocol [\[MS-RPCE\]](#).

Processing and Response Requirements:

- The behavior of this method is exactly same as that of [R_WinsStatus](#) except for the following:
 - There is no limit on the number of entries in the address-version map array.
 - This method SHOULD NOT be called with **Cmd_e** set to WINSINTF_E_ADDVERSMAP. If it is, the server returns an ERROR_WINS_INTERNAL error.
- Refer to R_WinsStatus and WINSINTF_RESULTS_NEW_T for the details of the behavior of this method.

3.1.4.21 R_WinsStatusWHdl (Opnum 20)

The R_WinsStatusWHdl method retrieves various configuration settings and the statistics of a **WINS server**.

```
DWORD R_WinsStatusWHdl(  
    [in, ref] WINSIF_HANDLE ServerHdl,  
    [in] WINSINTF_CMD_E Cmd_e,  
    [in, out, ref] PWINSINTF_RESULTS_NEW_T pResults  
);
```

ServerHdl: An **RPC** binding over IP address/HostName to the WINS server. RPC uses this binding internally to determine which WINS server the call is directed to. See [\[MSDN-Handles\]](#) for more information.

This value MUST be ignored by the WINS server on receipt.

Cmd_e: The command to be executed on the **target WINS server**.

pResults: A pointer to a structure of type [WINSINTF_RESULTS_NEW_T \(section 2.2.2.11\)](#) that contains the results of the command execution.

Return Values: A 32-bit unsigned integer that indicates the return status. A return value of ERROR_SUCCESS (0x00000000) indicates that operation completed successfully. A nonzero return value is a Win32 error code, as specified in [\[MS-ERREF\]](#). The following Win32 error codes can be returned:

Return value/code	Description
0x00000000 ERROR_SUCCESS	The call was successful.
0x00000FA0 ERROR_WINS_INTERNAL	An error occurred while processing the RPC call.
0x00000005 ERROR_ACCESS_DENIED	The caller doesn't have sufficient permissions.

The behavior of this method is the same as that of the **R_WinsStatusNew** method (section [3.1.4.20](#)).

3.1.4.22 R_WinsDoScavengingNew (Opnum 21)

The **R_WinsDoScavengingNew** method requests a specific **scavenging** operation on the **target WINS server**.

```
DWORD R_WinsDoScavengingNew(
    [in] handle_t ServerHdl,
    [in, ref] PWINSINTF_SCV_REQ_T pScvReq
);
```

ServerHdl: An **RPC** binding over IP address/HostName to the **WINS** server. RPC uses this binding internally to determine which WINS server the call is directed to.

pScvReq: A pointer to a WINSINTF_SCV_REQ_T structure (section [2.2.2.12](#)) that defines the type of scavenging operation.

Return Values: A 32-bit unsigned integer that indicates the return status. A return value of ERROR_SUCCESS (0x00000000) indicates that the operation completed successfully. A nonzero return value is a Win32 error code, as specified in [\[MS-ERREF\]](#). The following Win32 error codes can be returned:

Return value/code	Description
0x00000000 ERROR_SUCCESS	The call was successful.
0x00000005 ERROR_ACCESS_DENIED	The caller doesn't have sufficient permissions.

Exceptions Thrown: No exceptions SHOULD be thrown beyond those thrown by the underlying RPC protocol [\[MS-RPCE\]](#).

Processing and Response Requirements:

When [R_WinsDoScavengingNew](#) is called, the method returns immediately without waiting for scavenging to start. The server just queues a request for the scavenging operation; the internal state and configuration of the WINS server and the value of the *fForce* member in the WINSINTF_SCV_REQ_T structure (section 2.2.2.12) determine whether the scavenging occurs. Hence, callers to **R_WinsDoScavengingNew** SHOULD NOT treat a return code of ERROR_SUCCESS as indicating a successful scavenging operation. Instead, callers SHOULD rely on WINS event logs to determine whether or not the scavenging operation succeeded. The following table lists the events that indicate the status of scavenging.

Event ID	Event Name	Event Description
Informational events		
4143	WINS_EVT_SCV_RECS	WINS scavenged its records in the WINS database. The number of records scavenged is listed in the data section.
4144	WINS_EVT_SCV_RPLTOMB	WINS scavenged replica tombstones in the WINS database. The number of records Scavenged is in the data section.
4247	WINS_EVT_SCV_EXC	The WINS Scavenger thread encountered an error.
4250	WINS_EVT_SCV_ERR	The WINS Scavenger thread could not scavenge a record. This record is ignored and the Scavenger continues to the next available record. Check the application log for the Exchange component, ESENT.
4269	WINS_EVT_UNABLE_TO_CHG_PRIORITY	The Scavenger thread was unable to change its priority level.
4288	WINS_EVT_CLEANUP_OWNADDTBL_EXC	The Scavenger thread encountered an error while cleaning up the owner-address table. It will try again after the Verify interval has elapsed.
4328	WINS_EVT_ADMIN_SCVENGING_INITIATED	Administrator '%1' has initiated a scavenging operation.
4329	WINS_EVT_SCVENGING_STARTED	The WINS server has started a scavenging operation.
4330	WINS_EVT_SCVENGING_COMPLETED	The WINS server has completed the scavenging operation.
5001	WINS_EVT_SCV_RANGE	WINS is scavenging the locally owned records from the database. The version number range that is scavenged is in the data section, in the second to fifth words, in this order: from_version_number (low word, high word) to_version_number (low word, high word).
5002	WINS_EVT_SCV_CHUNK	WINS is scavenging a chunk of N records in the version number range from X to Y. N, X ,and Y (low word, high word for version numbers) are listed in the second to sixth words in the data section.
Warning events		
4150	WINS_EVT_ADJ_TIME_INTVL_R	WINS adjusted the scavenging related time interval, %1, so that it is compatible with the replication time intervals. The adjusted value for this scavenging parameter is given in the data section (second DWORD). This value was computed by WINS using an algorithm that MAY use the maximum replication time interval specified. The current value achieves a balance between consistency of databases across the network of WINS servers and the performance of the WINS servers.
4151	WINS_EVT_ADJ_TIME_INTVL	WINS adjusted the scavenging related time interval, %1. The adjusted value for this scavenging parameter is listed in the data section (second DWORD). This value was computed by WINS using an algorithm that tries to achieve a balance between consistency of databases across the network of WINS servers and the performance of the WINS servers.
4153	WINS_EVT_UNABLE_TO_VERIFY	The Scavenger thread found active replicas that needed

Event ID	Event Name	Event Description
		to be verified with the owner WINS server because they were older than the verify time interval. The table of owner-to-address mappings indicated the WINS was not active.
4164	WINS_EVT_FORCE_SCV_R_T	WINS was forced to scavenge replica tombstones of a WINS. The administrator on the computer forced the scavenging using winscl.exe. WINS does not scavenge replica tombstones unless they have timed out and the WINS has been running for at least three days. This ensures that the tombstones have replicated to other WINSes). In this case, the tombstones were timed out but the WINS had not been up for three days. The replica tombstones were deleted. This deletion does not constitute a problem unless the WINS servers are primary and backup to clients but not both Push and Pull partners of each other. If there are such WINSes, there is a low probability that this action will result in database inconsistency, but if it does, a consistent state can be achieved by initiating consistency checks using winscl.exe.

The following requirements and recommendations apply to a WINS server that processes a call to [R_WinsDoScavengingNew](#):

- Callers to R_WinsDoScavengingNew SHOULD have control level access. If an RPC client with a lower access level calls R_WinsDoScavengingNew, the server SHOULD return ERROR_ACCESS_DENIED.
- The WINS server queues a request on the target WINS server for the scavenging operation, and the method returns immediately with ERROR_SUCCESS as the status code.

3.1.5 Timer Events

No protocol timer events are required other than those in the underlying **RPC** protocol.

3.1.6 Other Local Events

No local events are maintained other than those in the underlying **RPC** protocol.

3.2 winsi2 Server Details

The methods supported by the **winsi2** interface are specified in [Message Processing Events and Sequencing Rules \(section 3.2.4\)](#).

3.2.1 Abstract Data Model

This section describes a conceptual model of possible data organization that an implementation can maintain to participate in this protocol. The described organization is provided to facilitate the explanation of how the protocol behaves. This document does not mandate that implementations adhere to this model as long as their external behavior is consistent with that described in this document.

A **NetBIOS name server (NBNS)** needs to maintain the following data structures:

Name record: A data structure that contains a name and the associated attributes.

Name records collection: A collection of all **name records** that are either registered by this NBNS server or obtained by replication.

Global version counter: A 64-bit unsigned integer that tracks the version number that is given to the next record to be updated.

Server configuration: Parameters maintained in persistent storage include the following:

- Refresh interval
- **Extinction interval**
- Extinction timeout
- Verify interval
- Process priority class
- Number of worker threads

3.2.2 Timers

No timers are required beyond those used internally by **RPC** to implement resiliency to network outages, as specified in [\[MS-RPCE\]](#) section 3.2.3.2.1.

3.2.3 Initialization

A **WINS** [winsi2](#) remote protocol server MUST be initialized by registering the **RPC** interface and listening on the dynamically allocated port assigned by RPC, as specified in section [2.1](#). The client MUST connect to a well-known RPC port on the WINS server to determine the endpoint of [winsi2](#). Before any client connection, the WINS server MUST wait for WINS [winsi2](#) to register with RPC before any clients can establish a connection.

3.2.4 Message Processing Events and Sequencing Rules

The [winsi2](#) interface provides methods that remotely configure, manage, and monitor a **WINS** server.

Methods in RPC Opnum Order

Method	Description
R_WinsTombstoneDbRecs	Tombstones a specified range of records belonging to a particular owner . Opnum: 0
R_WinsCheckAccess	Checks the granted level of access for the RPC caller. Opnum: 1

3.2.4.1 R_WinsTombstoneDbRecs (Opnum 0)

The [R_WinsTombstoneDbRecs](#) method **tombstones** records whose version numbers fall within a range of version numbers and are owned by a server with a specified address.

```
DWORD R_WinsTombstoneDbRecs(  
    [in] handle t ServerHdl,  
    [in, ref] PWINSINTF_ADD_T pWinsAdd,
```

```

[in] WINSINTF_VERS_NO_T MinVersNo,
[in] WINSINTF_VERS_NO_T MaxVersNo
);

```

ServerHdl: An **RPC** binding over IP address/HostName to the **WINS server**. RPC uses this binding internally to resolve which WINS server the call is directed to.

pWinsAdd: A pointer to the address of the **owner WINS server** whose records are to be tombstoned. This value MUST NOT be NULL.

MinVersNo: The lower bound on the range of version numbers that identifies the range of records to be tombstoned.

MaxVersNo: The upper bound on the range of version numbers that identifies the range of records to be tombstoned.

Return Values: A 32 bit unsigned integer value that indicates the return status. A return value of ERROR_SUCCESS (0x00000000) indicates that the operation completed successfully. Any other return value is a Win32 error code as specified in [\[MS-ERREF\]](#). The following Win32 error codes can be returned:

Return value/code	Description
0x00000000 ERROR_SUCCESS	The call was successful.
0x00000005 ERROR_ACCESS_DENIED	The caller does not have sufficient permissions.
0x00000FA0 ERROR_WINS_INTERNAL	An error occurred while processing the RPC call.

Exceptions Thrown: No exceptions SHOULD be thrown beyond those thrown by the underlying RPC protocol [\[MS-RPCE\]](#).

Processing and Response Requirements:

The following requirements and recommendations apply to a WINS server that processes a call to R_WinsTombstoneDbRecs:

- The R_WinsTombstoneDbRecs caller SHOULD have control-level access. If an RPC client with a lower access level calls this method, the server SHOULD return ERROR_ACCESS_DENIED.
- If the specified owner WINS server address is not found in the **owner-version map** table, the server SHOULD return ERROR_WINS_INTERNAL.
- If any error occurs during the retrieval or updating of database records, the server SHOULD return ERROR_WINS_INTERNAL.
- The server changes the state of the matching records to tombstoned. It also updates the version number and the ownership of these records so that the version number and record ownership are replicated on the partner WINS servers when replication takes place.
- The time stamp of the matching record is set to a string with the following format:

```
current time + tombstone timeout configured on the target WINS server
```

- If both *MinVersNo* and *MaxVersNo* are zero, all records matching the given **owner** address are tombstoned.

3.2.4.2 R_WinsCheckAccess (Opnum 1)

The *R_WinsCheckAccess* method retrieves the level of access the client is granted. [<12>](#)

```
DWORD R_WinsCheckAccess(  
    [in] handle_t ServerHdl,  
    [out] DWORD* Access  
);
```

ServerHdl: An **RPC** binding over IP address/HostName to the **WINS** server. RPC uses this binding internally to resolve which WINS server the call is directed to.

Access: Pointer to the access level value. This value MUST not be NULL. The following values are possible as output.

Name	Value
No access	0
Control level access	1
Query level access	2

Return Values: A 32-bit unsigned integer value that indicates the return status. A return value of *ERROR_SUCCESS* (0x00000000) indicates that the operation completed successfully. Any other return value is a Win32 error code as specified in [\[MS-ERREF\]](#). The following Win32 error codes can be returned:

Return value/code	Description
0x00000000 ERROR_SUCCESS	The call was successful.

Exceptions Thrown: No exceptions SHOULD be thrown beyond those thrown by the underlying RPC protocol [\[MS-RPCE\]](#).

Processing and Response Requirements:

Clients with any access level can call this method.

3.2.5 Timer Events

No protocol timer events are required other than those in the underlying **RPC** protocol.

3.2.6 Other Local Events

No local events are maintained other than those in the underlying **RPC** protocol.

4 Protocol Examples

4.1 Inserting a Record into a WINS Database

The following example illustrates the use of the **RPC** methods defined in this specification to insert a record into the database of a **WINS** server. If the WINS database on the specified server does not have a record with name "WINS-TEST-00001", then the client calls the RPC method [R_WinsRecordAction \(section 3.1.4.1\)](#) with the following parameters:

- *ServerHdl* set to the **endpoint** of the WINS server on which the *R_WinsRecordAction* method is executed.
- *ppRecAction* pointing to a structure of type [WINSINTF_RECORD_ACTION_T \(section 2.2.2.3\)](#), with members set as follows:
 - **Cmd_e** to **WINSINTF_E_INSERT**.
 - **pName** to point to the string "WINS-TEST-00001" followed by a **NetBIOS suffix** of 0x00.
 - **NameLen** to 16.
 - **TypOfRec_e** to 3, indicating a **multihomed (1)** record.
 - **NoOfAdds** to 2.
 - **pAdd** to point to two IP addresses: 192.168.1.1 and 192.168.1.2.
 - **NodeType** to 1, indicating a **p-node**.
 - **fStatic** to 0, indicating a **dynamic record**.

4.2 Releasing a Record from a WINS Database

The following example illustrates the use of the **RPC** methods defined in this specification to release a record from the database of a **WINS** server. If the WINS database on the specified server has a unique record with name "WINS-TEST-00001" mapped to the IP address 192.168.1.1, then the client calls the RPC method [R_WinsRecordAction \(section 3.1.4.1\)](#) with the following parameters:

- *ServerHdl* set to the endpoint of the WINS server on which the **R_WinsRecordAction** method is executed.
- *ppRecAction* pointing to a structure of type [WINSINTF_RECORD_ACTION_T \(section 2.2.2.3\)](#), with members set as follows:
 - **Cmd_e** to **WINSINTF_E_RELEASE**.
 - **pName** to point to the string "WINS-TEST-00001" followed by a **NetBIOS suffix** of 0x00.
 - **NameLen** to 16.
 - **TypOfRec_e** to 0.
 - **NoOfAdds** to 1.
 - **Add** to the IP address 192.168.1.1.

4.3 Deleting a Record from a WINS Database

The following example illustrates the use of the **RPC** methods defined in this specification to delete a record from the database of a **WINS** server. If the WINS database on the specified server has a **multihomed (1)** record with name "WINS-TEST-00001", the client calls the RPC method [R_WinsRecordAction \(section 3.1.4.1\)](#) with the following parameters:

- *ServerHdl* set to the endpoint of the WINS server on which the R_WinsRecordAction method is executed.
- *ppRecAction* pointing to a structure of type [WINSINTF_RECORD_ACTION_T \(section 2.2.2.3\)](#), with members set as follows:
 - **Cmd_e** to WINSINTF_E_DELETE.
 - **pName** to point to the string "WINS-TEST-00001" followed by a **NetBIOS suffix** of 0x00.
 - **NameLen** to 16.
 - **State_e** to 3 (DELETED).

4.4 Modifying a Record from a WINS Database

The following example illustrates the use of the **RPC** methods defined in this specification to modify a **WINS** server database record. If the WINS database on the specified server has a **multihomed (1) dynamic record** with the name "WINS-TEST-00001" and the node type set to **p-node**, the client calls the RPC method [R_WinsRecordAction \(section 3.1.4.1\)](#) with the following parameters:

- *ServerHdl* set to the endpoint of the WINS server on which the R_WinsRecordAction method is executed.
- *ppRecAction* pointing to a structure of type [WINSINTF_RECORD_ACTION_T \(section 2.2.2.3\)](#), with members set as follows:
 - **Cmd_e** to WINSINTF_E_MODIFY.
 - **pName** to point to the string "WINS-TEST-00001" followed by a **NetBIOS suffix** of 0x00.
 - **NameLen** to 16.
 - **TypOfRec_e** to 3, indicating a multihomed (1) record.
 - **NodeType** to 3.
 - **State_e** to 1.
 - **fStatic** to zero.

After executing the call to R_WinsRecordAction, the node type and the state of the existing record are modified to **h-node** and **RELEASED**, respectively.

4.5 Querying a Record from a WINS Database

The following example illustrates the use of the **RPC** methods defined in this specification to query a record from the database of a **WINS** server. This example assumes that the WINS database contained by the specified server has an active **multihomed (1) dynamic record** named "WINS-TEST-00001" mapped to two IP addresses: 192.168.1.1 and 192.168.1.2. IP address 192.168.1.1 has **node type p**, and its time stamp set to 0x61. IP address 192.168.1.2 has a version ID type and a time stamp set to 0x101E. The client calls the RPC method [R_WinsRecordAction \(section 3.1.4.1\)](#) with the following parameters:

- *ServerHdl* set to the endpoint of the WINS server on which the *R_WinsRecordAction* method is executed.
- *ppRecAction* pointing to a structure of type [WINSINTF_RECORD_ACTION_T \(section 2.2.2.3\)](#), with members set as follows:
 - **Cmd_e** to *WINSINTF_E_QUERY*.
 - **pName** to point to the string "WINS-TEST-00001" followed by a **NetBIOS suffix** of 0x00.
 - **NameLen** to 16.
- All other members are used for output, which the server assigns as follows:
 - **TypOfRec_e** contains 3.
 - **NoOfAdds** contains 2.
 - **pAdd** points to two IP addresses: 192.168.1.1 and 192.168.1.2.
 - **VersNo** contains 0x61.
 - **NodeType** contains 1.
 - **OwnerId** contains the **owner** IP address of the matching record.
 - **State_e** contains zero (**ACTIVE**).
 - **fStatic** contains zero.
 - **TimeStamp** contains 0x101E.

4.6 Retrieving All of the Records of a WINS Database

This example illustrates the use of the **RPC** methods defined in this specification to retrieve all the records from the database of a **WINS** server.

- The client calls the [R_WinsGetDbRecsByName](#) method repeatedly with the following parameters.
- Set *pWinsAdd* to NULL, *Location* to zero, *pName* to NULL, *NameLen* to zero, *fStaticOnly*, to 4 and *NoOfRecsDesired* to the desired number of records. As noted in the description of *R_WinsGetDbRecsByName*, the server resets the *NoOfRecsDesired* parameter to 5,000 if the parameter's value is greater than 5,000.
- Check how many *R_WinsGetDbRecsByName* has returned by looking at the value in the *NoOfRecs* field. If this value is less than the *NoOfRecsDesired* value, the retrieval is complete. Otherwise, if the number of returned values is the same as the value of *NoOfRecsDesired*, call to the *R_WinsGetDbRecsByName* with the following parameter settings:
- Set *pWinsAdd* to NULL, *Location* to 0, *pName* to the name of the last record retrieved in the previous iteration, *NameLen* to the length of *pName*, *fStaticOnly* to 4, and *NoOfRecsDesired* to the desired number of records.
- Repeated this procedure until the value of *NoOfRecs* is less than the value of *NoOfRecsDesired*.

4.7 Deleting All the Records of an Owner from a Particular WINS Server

This example illustrates the use of the **RPC** methods defined in this specification to delete all the records of an **owner** from the **target WINS server**.

The client calls the RPC method [R_WinsDelDbRecs](#) with the following parameters:

- The endpoint of the **WINS** server on which `R_WinsDelDbRecs` is executed (or from which the records are deleted as *ServerHdl*).
- Set *MinVersNo*, *MaxVersNo* and *pAdd->Type* to zero 0, *pAdd->Len* to 4, and *pAdd->IPAdd* to the IP address of the WINS server whose records are to be deleted.
- The successful completion of the `R_WinsDelDbRecs` call deletes all the records for an owner from the target WINS server database.

4.8 Deleting All the Records from a Particular WINS Server

This example illustrates the use of the **RPC** methods defined in this specification to delete all records from the **target WINS server**.

The client calls the RPC method [R_WinsStatusNew](#) with the following parameters:

- The endpoint of the **WINS** server on which the RPC method is executed, or from which the records are deleted, as *ServerHdl*.
- Set *Cmd_e* to `WINSINTF_E_CONFIG_ALL_MAPS`.
- The output of the call to `R_WinsStatusNew`, *pResults*, contains the list of owner addresses in the database of the target WINS server. For each owner address in *pResults->pAddVersMaps*, call the RPC method [R_WinsDelDbRecs \(section 3.1.4.9\)](#) by setting the parameters as follows:
 - Set *MinVersNo*, *MaxVersNo*, and *pAdd->Type* to 0.
 - Set *pAdd->Len* to 4.
 - Set *pAdd->IPAdd* to *pResults->pAddVersMaps[i]->Add*, where *i* denotes the *i*th iteration.

4.9 Triggering a Pull Replication Between Two WINS Servers

This example illustrates the use of the **RPC** methods defined in this specification to trigger a pull **replication** from one **WINS** server to another.

The client calls the RPC method [R_WinsTrigger](#) with the following parameters:

- Set the value of *ServerHdl* to the endpoint of the WINS server on which the pull replication is queued.
- Set *TrigType_e* and *pAdd->Type* to 0, *pAdd->Len* to 4 and *pAdd->IPAdd* to the IP address of the WINS server that serves as the partner for the pull replication.
- A return value of `ERROR_SUCCESS` means that the pull request has been queued successfully.

4.10 Backing Up a WINS Server Database

To back up the **WINS** server database, the client calls the **RPC** method [R_WinsBackup](#) with the following parameters:

- Set the value of *ServerHdl* to the endpoint of the WINS server on which the backup is performed.
- Set *pBackupPath* to the path on the server where the database is backed up.
- A return value of `ERROR_SUCCESS` indicates that the backup has been successful.

5 Security

5.1 Security Considerations for Implementers

RAIW allows any user to establish a connection to the **RPC** server. The protocol uses the underlying RPC protocol to retrieve the identity of the method caller as specified in [\[MS-RPCE\]](#). Clients create an authenticated RPC connection, and servers use this identity to perform specific access checks.

WINS server data and WINS server operations specified by this implementation are protected by access checks based on the identity of the RPC client.

Servers that implement this specification do not allow anonymous RPC connections and protect WINS access to all data and operations with access control checks based on client identity.

Clients or servers that implement this specification do not use RPC over named pipes because it is vulnerable to man-in-the-middle attacks. RPC over TCP/IP is used instead.

Servers that implement this protocol require clients to request `RPC_C_AUTHN_WINNT`, and servers enforce this requirement in order to protect the privacy of the communication with clients.

5.2 Index of Security Parameters

Security parameter	Section
<code>RPC_C_AUTHN_WINNT</code>	Section 2.1.1

6 Appendix A: Full IDL

6.1 Appendix A.1: winsif.idl

For ease of implementation, the full stand-alone **Interface Definition Language (IDL)** file for the [winsif](#) interface (section 3.1) is provided. Some of the data types and structures used by this protocol are defined in other documents. In order for this IDL to stand alone, the types and structures from [\[MS-DTYP\]](#) are imported.

```
import "ms-dtyp.idl";

#define WINSINTF_MAX_NO_RPL_PNRS 25

typedef PVOID LPVOID;
typedef LARGE_INTEGER WINSINTF_VERS_NO_T;

typedef struct _WINSINTF_ADD_T {
    BYTE Type;
    DWORD Len;
    DWORD IPAdd;
} WINSINTF_ADD_T, *PWINSINTF_ADD_T;

typedef enum _WINSINTF_PRIORITY_CLASS_E {
    WINSINTF_E_NORMAL = 0,
    WINSINTF_E_HIGH
} WINSINTF_PRIORITY_CLASS_E, *PWINSINTF_PRIORITY_CLASS_E;

typedef enum _WINSINTF_ACT_E {
    WINSINTF_E_INSERT = 0,
    WINSINTF_E_DELETE,
    WINSINTF_E_RELEASE,
    WINSINTF_E_MODIFY,
    WINSINTF_E_QUERY
} WINSINTF_ACT_E, *PWINSINTF_ACT_E;

typedef enum WINSINTF_TRIG_TYPE_E {    WINSINTF_E_PULL = 0,
    WINSINTF_E_PUSH,
    WINSINTF_E_PUSH_PROP
} WINSINTF_TRIG_TYPE_E, *PWINSINTF_TRIG_TYPE_E;

typedef struct WINSINTF_RECORD_ACTION_T {
    WINSINTF_ACT_E Cmd_e;
    [size is (NameLen + 1)] LPBYTE pName;
    DWORD NameLen;
    DWORD TypOfRec_e;
    DWORD NoOfAdds;
    [unique, size is (NoOfAdds)] PWINSINTF_ADD_T pAdd;
    WINSINTF_ADD_T Add;
    LARGE_INTEGER VersNo;
    BYTE NodeType;
    DWORD OwnerId;
    DWORD State_e;
    DWORD fStatic;
    DWORD_PTR TimeStamp;
} WINSINTF_RECORD_ACTION_T, *PWINSINTF_RECORD_ACTION_T;

typedef struct _WINSINTF_RPL_COUNTERS_T {
    WINSINTF_ADD_T Add;
    DWORD NoOfRpls;
    DWORD NoOfCommFails;
} WINSINTF_RPL_COUNTERS_T, *PWINSINTF_RPL_COUNTERS_T;

typedef struct _WINSINTF_STAT_T {
    struct {
        DWORD NoOfUniqueReg;
        DWORD NoOfGroupReg;
```

```

        DWORD NoOfQueries;
        DWORD NoOfSuccQueries;
        DWORD NoOfFailQueries;
        DWORD NoOfUniqueRef;
        DWORD NoOfGroupRef;
        DWORD NoOfRel;
        DWORD NoOfSuccRel;
        DWORD NoOfFailRel;
        DWORD NoOfUniqueCnf;
        DWORD NoOfGroupCnf;
    } Counters;
    struct {
        SYSTEMTIME WINSStartTime;
        SYSTEMTIME LastPScvTime;
        SYSTEMTIME LastATScvTime;
        SYSTEMTIME LastTombScvTime;
        SYSTEMTIME LastVerifyScvTime;
        SYSTEMTIME LastPRplTime;
        SYSTEMTIME LastATRplTime;
        SYSTEMTIME LastNTRplTime;
        SYSTEMTIME LastACTRplTime;
        SYSTEMTIME LastInitDbTime;
        SYSTEMTIME CounterResetTime;
    } TimeStamps;
    DWORD NoOfPnrs;
    [unique, size_is(NoOfPnrs)] PWINSINTF_RPL_COUNTERS_T pRplPnrs;
} WINSINTF_STAT_T, *PWINSINTF_STAT_T;

typedef struct WINSINTF_ADD_VERS_MAP_T {
    WINSINTF_ADD_T Add;
    LARGE_INTEGER VersNo;
} WINSINTF_ADD_VERS_MAP_T, *PWINSINTF_ADD_VERS_MAP_T;

typedef struct WINSINTF_RESULTS_T {
    DWORD NoOfOwners;
    WINSINTF_ADD_VERS_MAP_T AddVersMaps[WINSINTF_MAX_NO_RPL_PNRS];
    LARGE_INTEGER MyMaxVersNo;
    DWORD RefreshInterval;
    DWORD TombstoneInterval;
    DWORD TombstoneTimeout;
    DWORD VerifyInterval;
    DWORD WINSPriorityClass;
    DWORD NoOfWorkersThds;
    WINSINTF_STAT_T WINSStat;
} WINSINTF_RESULTS_T, *PWINSINTF_RESULTS_T;

typedef struct _WINSINTF_RESULTS_NEW_T {
    DWORD NoOfOwners;
    [unique, size_is(NoOfOwners)]
        PWINSINTF_ADD_VERS_MAP_T pAddVersMaps;

    LARGE_INTEGER MyMaxVersNo;
    DWORD RefreshInterval;
    DWORD TombstoneInterval;
    DWORD TombstoneTimeout;
    DWORD VerifyInterval;
    DWORD WINSPriorityClass;
    DWORD NoOfWorkersThds;
    WINSINTF_STAT_T WINSStat;
} WINSINTF_RESULTS_NEW_T, *PWINSINTF_RESULTS_NEW_T;

typedef enum _WINSINTF_CMD_E {
    WINSINTF_E_ADDVERSMAP = 0,
    WINSINTF_E_CONFIG,
    WINSINTF_E_STAT, WINSINTF_E_CONFIG_ALL_MAPS
} WINSINTF_CMD_E, *PWINSINTF_CMD_E;

typedef struct _WINSINTF_RECS_T {
    DWORD BuffSize;

```

```

        [unique, size is(NoOfRecs)] PWINSINTF_RECORD_ACTION_T pRow;
        DWORD NoOfRecs;
        DWORD TotalNoOfRecs;
    } WINSINTF_RECS_T, *PWINSINTF_RECS_T;

typedef struct WINSINTF_PULL_RANGE_INFO_T {
    LPVOID pPnr;
    WINSINTF_ADD_T OwnAdd;
    WINSINTF_VERS_NO_T MinVersNo;
    WINSINTF_VERS_NO_T MaxVersNo;
} WINSINTF_PULL_RANGE_INFO_T, *PWINSINTF_PULL_RANGE_INFO_T;

typedef struct _WINSINTF_BROWSER_INFO_T {
    DWORD dwNameLen;
    [string] LPBYTE pName;
} WINSINTF_BROWSER_INFO_T, *PWINSINTF_BROWSER_INFO_T;

typedef struct _WINSINTF_BROWSER_NAMES_T {
    DWORD EntriesRead;
    [unique, size is(EntriesRead)] PWINSINTF_BROWSER_INFO_T pInfo;
} WINSINTF_BROWSER_NAMES_T, *PWINSINTF_BROWSER_NAMES_T;

typedef enum WINSINTF_SCV_OPC_E {
    WINSINTF_E_SCV_GENERAL,
    WINSINTF_E_SCV_VERIFY
} WINSINTF_SCV_OPC_E, *PWINSINTF_SCV_OPC_E;

typedef struct _WINSINTF_SCV_REQ_T {
    WINSINTF_SCV_OPC_E Opcode e;
    DWORD Age;
    DWORD fForce;
} WINSINTF_SCV_REQ_T, *PWINSINTF_SCV_REQ_T;

typedef struct WINSINTF_BIND_DATA_T {
    DWORD fTcpIp;
    [string] LPSTR pServerAdd;
    [string] LPSTR pPipeName;
} WINSINTF_BIND_DATA_T, *PWINSINTF_BIND_DATA_T;

[
    uuid(45F52C28-7F9F-101A-B52B-08002B2EFABE),
    version(1.0),
    pointer_default(unique)
]

interface winsif {

#define MIDL_PASS

typedef [handle] PWINSINTF_BIND_DATA_T WINSIF_HANDLE;
typedef handle_t WINSIF2_HANDLE;

#define DECLARE_WINS_HANDLE( hdl ) [in] WINSIF2_HANDLE hdl,
#define DECLARE_WINS_HANDLE0( hdl ) [in] WINSIF2_HANDLE hdl

DWORD R_WinsRecordAction(
    DECLARE_WINS_HANDLE( ServerHdl )
    [in, out, ref] PWINSINTF_RECORD_ACTION_T *ppRecAction
);

DWORD R_WinsStatus(
    DECLARE_WINS_HANDLE( ServerHdl )
    [in] WINSINTF_CMD_E Cmd e,
    [in, out, ref] PWINSINTF_RESULTS_T pResults
);

DWORD R_WinsTrigger(
    DECLARE_WINS_HANDLE( ServerHdl )
    [in, ref] PWINSINTF_ADD_T pWinsAdd,

```

```

[in] WINSINTF TRIG TYPE E TrigType e
);

DWORD R_WinsDoStaticInit(
    DECLARE_WINS_HANDLE( ServerHdl )
    [in, unique, string] LPWSTR pDataFilePath,
    [in] DWORD fDel
);

DWORD R_WinsDoScavenging(
    DECLARE_WINS_HANDLE0( ServerHdl )
);

DWORD R_WinsGetDbRecs(
    DECLARE_WINS_HANDLE( ServerHdl )
    [in, ref] PWINSINTF_ADD_T pWinsAdd,
    [in] WINSINTF_VERS_NO_T MinVersNo,
    [in] WINSINTF_VERS_NO_T MaxVersNo,
    [out] PWINSINTF_RECS_T pRecs
);

DWORD R_WinsTerm(
    [in] handle t ServerHdl,
    [in] short fAbruptTem
);

DWORD R_WinsBackup(
    DECLARE_WINS_HANDLE( ServerHdl )
    [in, string, ref] LPBYTE pBackupPath,
    [in] short fIncremental
);

DWORD R_WinsDelDbRecs(
    DECLARE_WINS_HANDLE( ServerHdl )
    [in, ref] PWINSINTF_ADD_T pWinsAdd,
    [in] WINSINTF_VERS_NO_T MinVersNo,
    [in] WINSINTF_VERS_NO_T MaxVersNo
);

DWORD R_WinsPullRange(
    DECLARE_WINS_HANDLE( ServerHdl )
    [in, ref] PWINSINTF_ADD_T pWinsAdd,
    [in, ref] PWINSINTF_ADD_T pOwnerAdd,
    [in] WINSINTF_VERS_NO_T MinVersNo,
    [in] WINSINTF_VERS_NO_T MaxVersNo
);

DWORD R_WinsSetPriorityClass(
    DECLARE_WINS_HANDLE( ServerHdl )
    [in] WINSINTF_PRIORITY_CLASS_E PrCls_e
);

DWORD R_WinsResetCounters(
    DECLARE_WINS_HANDLE0( ServerHdl )
);

DWORD R_WinsWorkerThdUpd(
    DECLARE_WINS_HANDLE( ServerHdl )
    [in] DWORD NewNoOfNbtThds
);

DWORD R_WinsGetNameAndAdd(
    DECLARE_WINS_HANDLE( ServerHdl )
    [out, ref] PWINSINTF_ADD_T pWinsAdd,
    [out, string, size is(80)] LPBYTE pUncName
);

DWORD R_WinsGetBrowserNames_Old(
    DECLARE_WINS_HANDLE( ServerHdl )

```



```

        [out] PWINSINTF_BROWSER_NAMES_T pNames
    );

    DWORD R_WinsDeleteWins(
        DECLARE_WINS_HANDLE( ServerHdl )
        [in, ref] PWINSINTF_ADD_T pWinsAdd
    );

    DWORD R_WinsSetFlags(
        DECLARE_WINS_HANDLE( ServerHdl )
        [in]     DWORD fFlags
    );

    DWORD R_WinsGetBrowserNames(
        [in, ref] WINSIF_HANDLE ServerHdl,
        [out] PWINSINTF_BROWSER_NAMES_T pNames
    );

    DWORD R_WinsGetDbRecsByName(
        DECLARE_WINS_HANDLE( ServerHdl )
        [in, unique] PWINSINTF_ADD_T pWinsAdd,
        [in]     DWORD Location,
        [in, unique, size is(NameLen + 1)] LPBYTE pName,
        [in]     DWORD NameLen,
        [in]     DWORD NoOfRecsDesired,
        [in]     DWORD fOnlyStatic,
        [out] PWINSINTF_RECS_T pRecs
    );

    DWORD R_WinsStatusNew(
        DECLARE_WINS_HANDLE( ServerHdl )
        [in]     WINSINTF_CMD_E Cmd_e,
        [out] PWINSINTF_RESULTS_NEW_T pResults
    );

    DWORD R_WinsStatusWHdl(
        [in, ref] WINSIF_HANDLE ServerHdl,
        [in] WINSINTF_CMD_E Cmd_e,
        [in, out, ref] PWINSINTF_RESULTS_NEW_T pResults
    );

    DWORD R_WinsDoScavengingNew(
        DECLARE_WINS_HANDLE( ServerHdl )
        [in, ref] PWINSINTF_SCV_REQ_T pScvReq
    );
}

```

6.2 Appendix A.2: winsif2.idl

For ease of implementation, the full stand-alone **Interface Definition Language (IDL)** file for the [winsi2](#) interface (section 3.1) is provided. Some of the data types and structures used by this protocol are defined in other documents. In order for this IDL to stand alone, the types and structures from the [winsif](#) interface (section 6.1) IDL are imported.

```

import "ms-raiw winsif.idl";

[
    uuid(811109bf-a4e1-11d1-ab54-00a0c91e9b45),
    version(1.0),
    pointer_default(unique)
]

interface winsi2 {

```

```
#define MIDL_PASS

typedef handle_t WINSIF2_HANDLE;

DWORD
R_WinsTombstoneDbRecs(
    [in]          WINSIF2_HANDLE ServerHdl,
    [in, ref] PWINSINTF_ADD_T pWinsAdd,
    [in]          WINSINTF_VERS_NO_T MinVersNo,
    [in]          WINSINTF_VERS_NO_T MaxVersNo
);

DWORD
R_WinsCheckAccess(
    [in]          WINSIF2_HANDLE ServerHdl,
    [out]         DWORD *Access
);
}
```

7 Appendix B: Product Behavior

The information in this specification is applicable to the following Microsoft products or supplemental software. References to product versions include updates to those products.

The terms "earlier" and "later", when used with a product version, refer to either all preceding versions or all subsequent versions, respectively. The term "through" refers to the inclusive range of versions. Applicable Microsoft products are listed chronologically in this section.

Windows Client

- Windows Vista operating system
- Windows 7 operating system
- Windows 8 operating system
- Windows 8.1 operating system
- Windows 10 operating system
- Windows 11 operating system

Windows Server

- Windows Server 2003 operating system
- Windows Server 2008 operating system
- Windows Server 2008 R2 operating system
- Windows Server 2012 operating system
- Windows Server 2012 R2 operating system
- Windows Server 2016 operating system
- Windows Server operating system
- Windows Server 2019 operating system
- Windows Server 2022 operating system
- Windows Server 2025 operating system

Exceptions, if any, are noted in this section. If an update version, service pack or Knowledge Base (KB) number appears with a product name, the behavior changed in that update. The new behavior also applies to subsequent updates unless otherwise specified. If a product edition appears with the product version, behavior is different in that product edition.

Unless otherwise specified, any statement of optional behavior in this specification that is prescribed using the terms "SHOULD" or "SHOULD NOT" implies product behavior in accordance with the SHOULD or SHOULD NOT prescription. Unless otherwise specified, the term "MAY" implies that the product does not follow the prescription.

[<1> Section 2.1.1](#): Windows NT 4.0 operating system: The Remote Administrative Interface: WINS protocol does not define query-level access; the only access level available is control-level access. Clients that invoke **RPC** have control-level access.

<2> [Section 2.2.2.7](#): In Windows 2000 Server operating system and later if the value of the **WINSPriorityClass** member is other than NORMAL_PRIORITY_CLASS or HIGH_PRIORITY_CLASS, the system assumes the latter.

<3> [Section 3.1.4.1](#): Windows NT 4.0: The Remote Administrative Interface: WINS Protocol uses implicit binding, in which the RPC run-time library maintains the handle internally. No RPC methods except [R_WinsTombStoneDbRecs](#), [R_WinsTerm](#), and [R_WinsGetBrowserNames](#) take *ServerHdl* as a parameter.

<4> [Section 3.1.4.1](#): Windows NT 4.0: The RPC method caller is required to have control-level access regardless of the action used.

<5> [Section 3.1.4.1](#): In Windows 2000 Server and later, a maximum of 25 IP address mappings are allowed for a multihomed or special group **Name Record**.

<6> [Section 3.1.4.2](#): Windows NT 4.0: The [R_WinStatus](#) caller is required to have control-level access regardless of the command used.

<7> [Section 3.1.4.6](#): Windows NT 4.0: The RPC method caller is required to have control-level access regardless of the command used.

<8> [Section 3.1.4.10](#): The execution of [R_WinsPullRange](#) by a client with sufficient access permissions can cause the **WINS** service on the **target WINS server** to restart.

<9> [Section 3.1.4.14](#): Windows NT 4.0: The [R_WinsGetNameAndAdd](#) caller is required to have control-level access regardless of the command used.

<10> [Section 3.1.4.14](#): Windows NT 4.0: The [R_WinsGetNameAndAdd](#) caller does not need access permissions to call this method.

<11> [Section 3.1.4.19](#): Windows NT 4.0: The RPC method caller is required to have control-level access regardless of the command used.

<12> [Section 3.2.4.2](#): Windows NT 4.0: This RPC method is not supported.

8 Change Tracking

This section identifies changes that were made to this document since the last release. Changes are classified as Major, Minor, or None.

The revision class **Major** means that the technical content in the document was significantly revised. Major changes affect protocol interoperability or implementation. Examples of major changes are:

- A document revision that incorporates changes to interoperability requirements.
- A document revision that captures changes to protocol functionality.

The revision class **Minor** means that the meaning of the technical content was clarified. Minor changes do not affect protocol interoperability or implementation. Examples of minor changes are updates to clarify ambiguity at the sentence, paragraph, or table level.

The revision class **None** means that no new technical changes were introduced. Minor editorial and formatting changes may have been made, but the relevant technical content is identical to the last released version.

The changes made to this document are listed in the following table. For more information, please contact dochelp@microsoft.com.

Section	Description	Revision class
Z Appendix B: Product Behavior	Added Windows Server 2025 to the list of applicable products.	Major

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